

**Numerical Investigation of the Riemann  
Hypothesis  
via Dynamical Quantum Phase  
Transitions**

Implementation Plan and Theoretical Framework

March 31, 2026

<https://github.com/Protosophos/structure-space>

## Abstract

We present a detailed plan for the first public classical-computer implementation of the framework described by Wei et al. [1], which establishes an exact equivalence between the Riemann Hypothesis (RH) and the confinement of dynamical quantum phase transitions (DQPTs) to the critical line  $\text{Re}(s) = \frac{1}{2}$  in two carefully constructed quantum systems. Our implementation classically simulates both quantum systems—a probe spin coupled to a bath with logarithmic energy spectrum and a generalized Loschmidt echo construction—computing their respective order parameters on a fine grid in the  $(\beta, t)$  plane. The software is written in C++ with CUDA acceleration for large-scale parameter sweeps. By comparing detected phase-transition points against the known nontrivial zeros of the Riemann zeta function and scanning for off-critical-line zeros, we provide independent numerical evidence bearing on the Riemann Hypothesis through the lens of quantum many-body physics.

We emphasize that this numerical approach cannot prove the Riemann Hypothesis; it can, however, either discover a counterexample (disproving RH) or provide additional computational evidence consistent with RH, while simultaneously validating the Wei et al. theoretical framework [1] against established zero tables.

**Notation.** Throughout this document,  $\log$  denotes the natural logarithm (base  $e$ ). We write  $\log_2$  explicitly when base-2 logarithm is intended.

# Contents

|          |                                                                    |           |
|----------|--------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction and Motivation</b>                                 | <b>1</b>  |
| 1.1      | The Riemann Hypothesis . . . . .                                   | 1         |
| 1.2      | Computational Approaches to RH . . . . .                           | 1         |
| 1.3      | The Hilbert–Pólya Conjecture and DQPTs . . . . .                   | 2         |
| 1.4      | Scope and Limitations of the DQPT Reformulation . . . . .          | 3         |
| <b>2</b> | <b>Theoretical Framework</b>                                       | <b>3</b>  |
| 2.1      | Dynamical Quantum Phase Transitions . . . . .                      | 3         |
| 2.2      | The Wei et al. Construction . . . . .                              | 4         |
| 2.2.1    | System 1: Probe Spin Coherence with Logarithmic Spectrum . . . . . | 4         |
| 2.2.2    | System 2: Generalized Loschmidt Amplitude . . . . .                | 11        |
| 2.3      | The RH–DQPT Equivalence . . . . .                                  | 12        |
| 2.4      | Free Energy and Singularities . . . . .                            | 13        |
| <b>3</b> | <b>Numerical Methods</b>                                           | <b>14</b> |
| 3.1      | Computation of Partial Dirichlet Series . . . . .                  | 14        |
| 3.2      | Riemann–Siegel Theta Function . . . . .                            | 18        |
| 3.2.1    | The Riemann–Siegel Formula . . . . .                               | 19        |
| 3.3      | Zero Detection . . . . .                                           | 21        |
| 3.3.1    | Bisection on the Hardy $Z$ -function . . . . .                     | 21        |
| 3.3.2    | Gram’s Law and Its Failures . . . . .                              | 21        |
| 3.3.3    | Turing’s Method for Zero Count Certification . . . . .             | 21        |
| 3.3.4    | Expected Zero Density . . . . .                                    | 22        |
| 3.4      | Convergence Analysis . . . . .                                     | 22        |
| 3.4.1    | Convergence of $L(\beta, t)$ with $N$ . . . . .                    | 22        |
| 3.4.2    | Required $N$ as a Function of $t$ and Precision . . . . .          | 23        |
| 3.4.3    | Computational Complexity . . . . .                                 | 23        |
| 3.5      | GPU Parallelization Strategy . . . . .                             | 24        |
| 3.5.1    | Parallelism Structure . . . . .                                    | 24        |
| 3.5.2    | Memory Model . . . . .                                             | 24        |
| 3.5.3    | Expected Throughput . . . . .                                      | 24        |
| 3.5.4    | Batch Size Estimation . . . . .                                    | 25        |
| <b>4</b> | <b>Experimental Design</b>                                         | <b>26</b> |
| 4.1      | Hypothesis . . . . .                                               | 26        |
| 4.2      | Test Protocol . . . . .                                            | 26        |
| 4.3      | Parameters . . . . .                                               | 27        |
| 4.4      | Validation . . . . .                                               | 29        |

|           |                                                      |           |
|-----------|------------------------------------------------------|-----------|
| 4.4.1     | Unit Tests Against Reference Values                  | 29        |
| 4.4.2     | Cross-Validation: System 1 vs System 2               | 29        |
| 4.4.3     | Convergence Tests                                    | 29        |
| 4.4.4     | Comparison with Odlyzko's Tables                     | 29        |
| 4.5       | Controls                                             | 29        |
| <b>5</b>  | <b>Error Analysis</b>                                | <b>30</b> |
| 5.1       | Truncation Error (Finite $N$ )                       | 30        |
| 5.2       | Floating-Point Precision                             | 31        |
| 5.3       | Underflow Guard for $\log  L_N $                     | 32        |
| 5.4       | Grid Discretization Error                            | 33        |
| 5.5       | False Positive and False Negative Rates              | 33        |
| <b>6</b>  | <b>Expected Results</b>                              | <b>33</b> |
| 6.1       | If RH Holds (Expected Scenario)                      | 33        |
| 6.2       | If RH Fails (Unexpected Scenario)                    | 34        |
| 6.3       | Limitations                                          | 35        |
| <b>7</b>  | <b>Relation to Prior Work</b>                        | <b>35</b> |
| 7.1       | Odlyzko's Numerical Verification                     | 35        |
| 7.2       | Berry–Keating Conjecture                             | 36        |
| 7.3       | Connes' Operator-Algebraic Approach                  | 36        |
| 7.4       | Guth–Maynard 2024 Breakthrough                       | 37        |
| 7.5       | How Our Implementation Differs                       | 37        |
| <b>8</b>  | <b>Directions Toward Proof</b>                       | <b>37</b> |
| 8.1       | Lee–Yang Mechanism for Alternating Series            | 38        |
| 8.2       | Energy–Entropy Balance at $\beta = 1/2$              | 38        |
| 8.3       | Universality of the DQPT Critical Point              | 39        |
| 8.4       | Selberg Trace Formula Connection                     | 40        |
| 8.5       | Random Matrix Universality from the DQPT Perspective | 40        |
| <b>9</b>  | <b>Implementation Architecture</b>                   | <b>41</b> |
| 9.1       | File Structure                                       | 41        |
| 9.2       | Language Choices                                     | 42        |
| 9.3       | Dependencies                                         | 43        |
| 9.4       | Build System                                         | 43        |
| <b>10</b> | <b>Success Criteria (Quantitative)</b>               | <b>43</b> |
| 10.1      | Verification Criteria (Must Pass)                    | 44        |
| 10.2      | RH-Relevant Criteria                                 | 45        |

---

|                                   |           |
|-----------------------------------|-----------|
| <b>11 Timeline and Milestones</b> | <b>45</b> |
| <b>References</b>                 | <b>47</b> |

# 1 Introduction and Motivation

## 1.1 The Riemann Hypothesis

The Riemann zeta function is defined for  $\operatorname{Re}(s) > 1$  by the absolutely convergent series

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \quad (1.1)$$

and extended to the entire complex plane (except for a simple pole at  $s = 1$ ) by analytic continuation. The functional equation

$$\zeta(s) = \chi(s) \zeta(1-s), \quad \chi(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s), \quad (1.2)$$

equivalently expressed using the completed zeta function

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s), \quad (1.3)$$

which satisfies the symmetric form  $\xi(s) = \xi(1-s)$ , implies that  $\zeta(s)$  vanishes at  $s = -2, -4, -6, \dots$  (the *trivial zeros*). The Riemann Hypothesis [2] asserts:

**RH.** *All nontrivial zeros of  $\zeta(s)$  have real part equal to  $\frac{1}{2}$ .*

Equivalently, if  $\zeta(\sigma + it) = 0$  with  $0 < \sigma < 1$  and  $t \in \mathbb{R}$ , then  $\sigma = \frac{1}{2}$ . This remains the most important open problem in mathematics, with profound implications for the distribution of prime numbers, the error term in the prime counting function, and numerous results across number theory, analysis, and mathematical physics that are conditional on RH.

## 1.2 Computational Approaches to RH

The dominant computational approach to RH has been direct numerical verification: computing  $\zeta(\frac{1}{2} + it)$  along the critical line and certifying that all zeros up to some height  $T$  lie on it. Odlyzko [3, 4] computed specific high zeros of  $\zeta$ —around the  $10^{20}$ th and  $10^{22}$ nd zeros—and large blocks of consecutive zeros near these heights, providing detailed statistical data on zero spacing and pair correlation. Gourdon [5] systematically verified that all of the first  $10^{13}$  nontrivial zeros lie on the critical line, using the Odlyzko–Schönhage algorithm based on the Riemann–Siegel formula. These computations exploit highly optimized algorithms specific to the zeta function and achieve extraordinary precision.

Our approach differs fundamentally. Rather than computing  $\zeta$  directly, we *classically* evaluate the mathematical expressions (partial Dirichlet sums, Loschmidt

amplitudes, rate functions) that arise from quantum-mechanical systems whose observable phase transitions correspond exactly to the zeros of  $\zeta$ .<sup>1</sup> This provides:

1. An independent computational pathway to zero detection, serving as a cross-check.
2. Physical intuition about the structure of zeta zeros through the language of quantum phase transitions.
3. A connection (though not a direct realization) to the Hilbert–Pólya program<sup>2</sup>, linking RH to quantum dynamical phenomena rather than to a single spectral operator.

### 1.3 The Hilbert–Pólya Conjecture and DQPTs

Berry and Keating [6] proposed that the Hilbert–Pólya operator might be related to the quantization of the classical Hamiltonian  $H = xp$ , connecting RH to quantum chaos.

The Wei et al. construction [1] takes a different but related path. Instead of seeking a single operator whose spectrum encodes the zeros, they construct quantum systems whose *dynamical* phase transitions—non-analyticities in a rate function as a function of real time—occur precisely at the zeta zeros. The key insight is that the Loschmidt amplitude of these systems, when evaluated at inverse temperature  $\beta$ , is proportional to the Dirichlet eta function  $\eta(\beta + it)$  (System 1) or the Hardy  $Z$ -function (System 2). The RH is then equivalent to the statement that DQPTs occur only at  $\beta = \frac{1}{2}$ .

This reformulation shifts RH from a statement about complex analysis to one about quantum statistical mechanics, opening new avenues for both theoretical and numerical investigation.

---

<sup>1</sup>To be precise: no quantum hardware or quantum simulation is involved. The computation reduces to evaluating partial sums of Dirichlet series and related functions on a classical computer. The quantum-mechanical language (Hamiltonians, coherence functions, Loschmidt echoes) provides the *framework* and *interpretation*, not the computational method. The value of this framing lies in connecting the Riemann Hypothesis to the well-developed theory of dynamical quantum phase transitions, not in any computational advantage over direct zeta evaluation.

<sup>2</sup>The Hilbert–Pólya conjecture, traditionally attributed to c. 1914 (though the earliest written record is later), posits the existence of a self-adjoint operator whose eigenvalues are the imaginary parts of the nontrivial zeta zeros. If such an operator exists, the reality of its eigenvalues would immediately imply RH.

## 1.4 Scope and Limitations of the DQPT Reformulation

We emphasize an important conceptual point: the DQPT–RH equivalence established by Wei et al. is a *reformulation* of the Riemann Hypothesis in the language of quantum statistical mechanics, not a proof mechanism. The equivalence states that RH holds if and only if DQPTs are confined to  $\beta = \frac{1}{2}$ , but this reformulation does not by itself provide any new leverage toward proving (or disproving) RH. The difficulty of the problem is preserved: it is merely re-expressed in physical language.

In particular:

1. The quantum systems (System 1 and System 2) are *constructed* so that their partition functions and order parameters reproduce the Dirichlet eta function and Hardy  $Z$ -function. The physics does not generate RH as a consequence of some independent physical principle; rather, the mathematics of RH is *encoded* into the physics by design.
2. The numerical investigation described in this document can provide computational evidence consistent with (or inconsistent with) RH, but it cannot prove RH. This is a fundamental limitation shared by all numerical approaches.
3. The value of the DQPT framework lies in (a) providing physical intuition about the structure of zeta zeros, (b) connecting RH to the well-developed theory of quantum phase transitions, and (c) potentially suggesting new analytical approaches inspired by the physics (see Section 8).

## 2 Theoretical Framework

### 2.1 Dynamical Quantum Phase Transitions

Dynamical quantum phase transitions (DQPTs) were introduced by Heyl, Polkovnikov, and Kehrein [7] as real-time analogs of equilibrium phase transitions. The central object is the Loschmidt amplitude:

**Definition 2.1** (Loschmidt amplitude). *Given an initial state  $|\psi_0\rangle$  (typically the ground state of a pre-quench Hamiltonian  $H_0$ ) and a post-quench Hamiltonian  $H$ , the Loschmidt amplitude is*

$$\mathcal{G}(t) = \langle \psi_0 | e^{-iHt} | \psi_0 \rangle. \quad (2.1)$$

*The return rate (or rate function) is defined as*

$$g(t) = -\frac{1}{N} \ln |\mathcal{G}(t)|^2, \quad (2.2)$$

where  $N$  is the system size. A DQPT occurs at time  $t^*$  if  $g(t)$  is non-analytic at  $t = t^*$ .

The connection to equilibrium statistical mechanics arises through the formal correspondence  $t \rightarrow -i\beta$  (Wick rotation), under which the Loschmidt amplitude becomes

$$\mathcal{G}(-i\beta) = \langle \psi_0 | e^{-\beta H} | \psi_0 \rangle. \quad (2.3)$$

For the specific case where  $|\psi_0\rangle$  is a uniform superposition over energy eigenstates, this equals  $Z(\beta)/\dim(\mathcal{H})$ . In general, the relation depends on the overlap of  $|\psi_0\rangle$  with the eigenstates of  $H$ .

Non-analyticities in the free energy density  $f(\beta) = -(1/N) \ln Z(\beta)$  as a function of inverse temperature correspond to equilibrium phase transitions. By the Lee–Yang theorem [8], these non-analyticities are controlled by the zeros of the partition function in the complex  $\beta$ -plane. Analogously, DQPTs are controlled by the zeros of the Loschmidt amplitude in the complex time plane. When these zeros cross the real time axis, the rate function  $g(t)$  exhibits non-analytic behavior (cusps, divergences, or discontinuities in derivatives), signaling a DQPT [7].

**Connection to Lee–Yang zeros.** In the Lee–Yang theory [8, 9], equilibrium phase transitions arise when zeros of the partition function  $Z(\beta, h)$ —viewed as a function of complex fugacity or magnetic field—pinch the real axis in the thermodynamic limit. The density of these zeros determines the order and nature of the transition. In the DQPT framework, the analogous role is played by the Fisher zeros of the Loschmidt amplitude in the complex time plane. The Wei et al. construction arranges for these Fisher zeros to coincide with the nontrivial zeros of the Riemann zeta function.

## 2.2 The Wei et al. Construction

### 2.2.1 System 1: Probe Spin Coherence with Logarithmic Spectrum

**Setup.** Consider a composite system consisting of a probe spin- $\frac{1}{2}$  coupled to a bath of  $N$  quantum levels. The bath Hamiltonian has a logarithmic energy spectrum:

$$H_0 = \sum_{n=1}^N \ln(n) |n\rangle\langle n|, \quad (2.4)$$

where  $\{|n\rangle\}_{n=1}^N$  is an orthonormal basis for the bath Hilbert space.

*Remark 2.2* (On the spectrum). The logarithmic spectrum  $E_n = \ln(n)$  is reverse-engineered to produce the Dirichlet series in the partition function; it is not a spectrum that arises naturally in a standard condensed-matter or atomic physics

context. However, logarithmic spectra do arise in the theory of quantum mechanics on hyperbolic surfaces: the Selberg zeta function

$$Z_\Gamma(s) = \prod_{\{p\}} \prod_{k=0}^{\infty} (1 - e^{-(s+k)\ell_p}), \quad (2.5)$$

associated to a discrete subgroup  $\Gamma$  of  $\mathrm{SL}(2, \mathbb{R})$ , encodes lengths  $\ell_p$  of primitive closed geodesics, and the corresponding quantum Hamiltonian (the Laplace–Beltrami operator on the quotient surface) has a spectral theory intimately related to zeta functions [10]. This provides a potential natural realization of the logarithmic spectrum, though the precise connection to the Wei et al. construction remains to be established.

The probe spin couples to the bath such that the effective Hamiltonian for the coupled evolution is

$$H = \sigma_z \otimes H_0, \quad (2.6)$$

where  $\sigma_z$  is the Pauli- $Z$  operator acting on the probe spin.

**Thermal state preparation.** The bath is prepared in a thermal (Gibbs) state at inverse temperature  $\beta$ :

$$\rho_{\text{bath}}(\beta) = \frac{e^{-\beta H_0}}{Z(\beta, H_0)}, \quad (2.7)$$

where the partition function is

$$Z(\beta, H_0) = \mathrm{Tr}[e^{-\beta H_0}] = \sum_{n=1}^N e^{-\beta \ln(n)} = \sum_{n=1}^N n^{-\beta}. \quad (2.8)$$

Note that  $Z(\beta, H_0)$  is precisely the partial zeta function. In the thermodynamic limit  $N \rightarrow \infty$  and for  $\beta > 1$ , this converges to  $\zeta(\beta)$  [11].

The probe spin is prepared in the state  $|+\rangle = (|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$ .

**Time evolution.** The probe-bath system is prepared in the product state

$$\rho_{\text{total}} = |+\rangle\langle+| \otimes \rho_{\text{bath}}(\beta), \quad (2.9)$$

where  $|+\rangle\langle+|$  is the pure state projector for the probe spin and  $\rho_{\text{bath}}(\beta)$  is the thermal state defined above. Under the coupled Hamiltonian  $H = \sigma_z \otimes H_0$ , the total state evolves as

$$\rho_{\text{total}}(t) = e^{-iHt} \rho_{\text{total}} e^{iHt}. \quad (2.10)$$

The reduced density matrix of the probe spin, obtained by tracing over the bath (partial trace), yields the coherence (off-diagonal element):

$$\langle\sigma_+\rangle(t) = \mathrm{Tr}_{\text{bath}}[\rho_{\text{bath}} e^{2iH_0t}]. \quad (2.11)$$

**Derivation of the accumulated phase factor.** We compute:

$$\begin{aligned}
\langle \sigma_+ \rangle(t) &= \text{Tr}[\rho_{\text{bath}} e^{2iH_0 t}] \\
&= \frac{1}{Z(\beta)} \sum_{n=1}^N e^{-\beta \ln(n)} e^{2i \ln(n) t} \\
&= \frac{1}{Z(\beta)} \sum_{n=1}^N n^{-\beta} n^{2it} \\
&= \frac{1}{Z(\beta)} \sum_{n=1}^N n^{-(\beta - 2it)} \\
&= \frac{S_N(\beta - 2it)}{Z(\beta)}, \tag{2.12}
\end{aligned}$$

where  $S_N(s) = \sum_{n=1}^N n^{-s}$  is the partial Dirichlet series.

Note that the coherence  $\langle \sigma_+ \rangle$  involves  $S_N(\beta - 2it)$ , not  $S_N(\beta + it)$ . Wei et al. [1] introduce three modifications to obtain the accumulated phase factor  $\mathcal{L}(\beta, t)$ :

- (a) A reparametrization  $t \rightarrow t/2$  absorbs the factor of 2: replacing  $2t$  by  $t$  in the exponent gives  $S_N(\beta - it)$ . This is purely a rescaling of the time variable and does not affect the location of zeros (a zero of  $S_N(\beta - it_0)$  at time  $t_0$  corresponds to a zero of  $S_N(\beta - 2it'_0)$  at time  $t'_0 = t_0/2$ ).
- (b) A sign convention: the coherence involves  $n^{-(\beta - it)} = n^{-\beta} n^{it}$ , while the standard Dirichlet series uses  $n^{-(\beta + it)} = n^{-\beta} n^{-it}$ . The passage from  $-it$  to  $+it$  corresponds to complex conjugation of the time evolution phase, equivalent to choosing the opposite sign convention for the coupling Hamiltonian. This does not affect the zero locations (if  $\eta(\beta + it_0) = 0$ , then  $\eta(\beta - it_0) = 0$  by the reflection principle, since the eta coefficients are real).
- (c) An alternating sign  $(-1)^{n+1}$  is introduced, corresponding physically to a staggered coupling between the probe and even/odd bath levels. This converts the ordinary Dirichlet series into the alternating Dirichlet eta series.<sup>3</sup>

With these three modifications, the accumulated phase factor becomes:

$$\mathcal{L}(\beta, t) = \frac{1}{Z(\beta)} \sum_{n=1}^N (-1)^{n+1} n^{-(\beta + it)}. \tag{2.13}$$

---

<sup>3</sup>This staggered coupling requires that the probe spin distinguishes the parity of the energy index  $n$  and reverses its coupling accordingly. Since  $n$  is an energy index rather than a spatial coordinate, this coupling is *non-local in energy space* and does not arise naturally in any known many-body system. Together with the logarithmic spectrum  $E_n = \ln n$ , this reinforces the point made in Section 1.4: the quantum system is *engineered* to reproduce the Dirichlet eta function, and the “physics” is a mathematical reformulation rather than an independent physical mechanism.

The factor  $(-1)^{n+1}$  converts the ordinary Dirichlet series into the alternating Dirichlet series, which defines the Dirichlet eta function:

$$\eta(s) = \sum_{n=1}^{\infty} (-1)^{n+1} n^{-s} = (1 - 2^{1-s}) \zeta(s). \quad (2.14)$$

**Why  $\mathcal{L}$  vanishes at zeta zeros.** We now prove the key result. In the thermodynamic limit  $N \rightarrow \infty$ :

**Step 1.** The numerator of  $\mathcal{L}(\beta, t)$  converges:

$$\sum_{n=1}^{\infty} (-1)^{n+1} n^{-(\beta+it)} = \eta(\beta + it). \quad (2.15)$$

This series converges for  $\text{Re}(\beta + it) = \beta > 0$  by Dirichlet's test (Abel summation): the partial sums of  $(-1)^{n+1}$  are bounded and  $n^{-\beta}$  decreases monotonically to zero [11]. (Note: the classical Leibniz alternating series test applies only to real series; for complex  $s = \beta + it$  with  $t \neq 0$ , convergence follows from Dirichlet's test.)

**Step 2.** The Dirichlet eta function satisfies the identity

$$\eta(s) = (1 - 2^{1-s}) \zeta(s) \quad (2.16)$$

for all  $s \in \mathbb{C}$  (by analytic continuation). This identity is proved by observing that for  $\text{Re}(s) > 1$ :

$$\zeta(s) - \eta(s) = 2 \sum_{n=1}^{\infty} (2n)^{-s} = 2^{1-s} \zeta(s), \quad (2.17)$$

whence  $\eta(s) = \zeta(s) - 2^{1-s} \zeta(s) = (1 - 2^{1-s}) \zeta(s)$ . Both sides are meromorphic on  $\mathbb{C}$  and agree on  $\text{Re}(s) > 1$ ; by the identity theorem for meromorphic functions, they agree everywhere on  $\mathbb{C}$ .

**Step 3.** The prefactor  $(1 - 2^{1-s})$  vanishes when  $2^{1-s} = 1$ , i.e., when  $(1 - s) \ln 2 = 2\pi i k$  for some integer  $k$ . Writing  $s = \beta + it$ :

$$(1 - \beta - it) \ln 2 = 2\pi i k. \quad (2.18)$$

Separating real and imaginary parts:

$$\text{Real: } (1 - \beta) \ln 2 = 0, \quad \text{hence } \beta = 1. \quad (2.19)$$

$$\text{Imaginary: } -t \ln 2 = 2\pi k, \quad \text{hence } t = -2\pi k / \ln 2 = 2\pi(-k) / \ln 2. \quad (2.20)$$

Since  $k$  ranges over all integers, the set of solutions is  $t = 2\pi m / \ln 2$  for  $m \in \mathbb{Z}$  (relabeling  $m = -k$ ). The prefactor zeros thus lie at  $s = 1 + i(2\pi m / \ln 2)$  for  $m \in \mathbb{Z}$ . Since  $\beta = 1$  lies *outside* the open critical strip  $0 < \beta < 1$  (and no nontrivial zeta

zero has  $\text{Re}(s) = 1$  by the prime number theorem [11]), these prefactor zeros never coincide with nontrivial zeta zeros.

**Step 4.** Therefore, for  $s = \beta + it$  with  $0 < \beta < 1$ :

$$\eta(s) = 0 \iff \zeta(s) = 0. \quad (2.21)$$

**Step 5: The partition function divergence.** The partition function  $Z(\beta) = \sum_{n=1}^N n^{-\beta}$  converges to  $\zeta(\beta)$  only for  $\beta > 1$ . For  $\beta = \frac{1}{2}$  (the critical case for RH),  $Z(\frac{1}{2}) = \sum 1/\sqrt{n}$  diverges as  $2\sqrt{N}$  by the integral test [11].

This means the naive ratio  $\mathcal{L} = \eta/Z$  is a  $0/\infty$  form at the zeta zeros when  $\beta = \frac{1}{2}$ . We must handle this carefully.

Following Wei et al. [1], the physically meaningful quantity is the *finite-N* expression:

$$\mathcal{L}_N(\beta, t) = \frac{\sum_{n=1}^N (-1)^{n+1} n^{-(\beta+it)}}{\sum_{n=1}^N n^{-\beta}}. \quad (2.22)$$

For finite  $N$ , both numerator and denominator are finite. The key observation is that the *rate function* (free energy density) has a well-defined thermodynamic limit:

$$\mathcal{F}_1(\beta, t) = - \lim_{N \rightarrow \infty} \frac{1}{\ln N} \ln |\mathcal{L}_N(\beta, t)|. \quad (2.23)$$

To see this, note that for the numerator:

- At a zeta zero  $s_0 = \beta_0 + it_0$ : the partial eta sum  $S_N^\eta(s_0) = O(N^{-\beta_0})$  by the Dirichlet test remainder bound [11].
- Away from zeros:  $S_N^\eta(s) \rightarrow \eta(s) \neq 0$ , a nonzero constant.

For the denominator  $Z_N(\beta)$ :

- For  $0 < \beta < 1$ :  $Z_N(\beta) \sim N^{1-\beta}/(1-\beta)$  by Euler–Maclaurin summation [11].

Therefore:

$$|\mathcal{L}_N(\beta, t)| \sim |S_N^\eta(s)| \frac{1-\beta}{N^{1-\beta}}. \quad (2.24)$$

At a zero ( $s = s_0$ ):  $|S_N^\eta(s_0)| = O(N^{-\beta_0})$ , so  $|\mathcal{L}_N| \sim N^{\beta-\beta_0-1}$ . Away from a zero:  $|S_N^\eta(s)| = O(1)$ , so  $|\mathcal{L}_N| \sim N^{\beta-1}$ .

We verify the limit (2.23) exists. Away from zeros,  $\ln |\mathcal{L}_N| = \ln |\eta(s)(1-\beta)| - (1-\beta) \ln N$ , giving:

$$\begin{aligned} \mathcal{F}_1 &= - \lim_{N \rightarrow \infty} \frac{1}{\ln N} [\ln |\eta(s)(1-\beta)| - (1-\beta) \ln N] \\ &= - \lim_{N \rightarrow \infty} \frac{\ln |\eta(s)(1-\beta)|}{\ln N} + (1-\beta) = 1-\beta. \end{aligned} \quad (2.25)$$

At a zero,  $\ln |\mathcal{L}_N| = \ln D + (\beta - \beta_0 - 1) \ln N$ :

$$\begin{aligned}\mathcal{F}_1 &= - \lim_{N \rightarrow \infty} \frac{1}{\ln N} [\ln D + (\beta - \beta_0 - 1) \ln N] \\ &= \beta_0 + 1 - \beta.\end{aligned}\tag{2.26}$$

Both limits exist and are finite. This confirms that  $\mathcal{F}_1$  is well-defined despite the divergence of  $Z(\beta)$  for  $\beta \leq 1$ .

**Step 6: Equivalence statement.** The Riemann Hypothesis is equivalent to:

$$\text{For all } t \in \mathbb{R} : \mathcal{F}_1(\beta, t) \text{ exhibits a singularity (DQPT) only at } \beta = \tfrac{1}{2}.\tag{2.27}$$

A counterexample to RH—a zero  $\rho = \beta_0 + it_0$  with  $\beta_0 \neq \frac{1}{2}$ —would manifest as a singularity in  $\mathcal{F}_1$  at  $\beta = \beta_0$ . This is numerically detectable by scanning  $\mathcal{F}_1(\beta, t)$  over a grid in the  $(\beta, t)$  plane and looking for anomalous values of  $\mathcal{F}_1$  at  $\beta \neq \frac{1}{2}$ .

**Convergence of the eta partial sums.** The Dirichlet eta series converges for  $\text{Re}(s) > 0$  by Dirichlet’s test (Abel summation), but only conditionally (not absolutely) for  $0 < \text{Re}(s) \leq 1$ . The convergence rate is:

$$|\eta(s) - \eta_N(s)| \leq C(s) (N + 1)^{-\beta},\tag{2.28}$$

where  $C(s) = O(1)$  for  $s$  bounded away from  $\text{Re}(s) = 0$ , but  $C(s)$  grows as  $|\text{Im}(s)|$  increases for fixed  $\text{Re}(s)$  near 0. This is  $O(N^{-1/2})$  at  $\beta = \frac{1}{2}$ , requiring  $N \sim 10^{2k}$  to achieve  $k$  digits of precision. For double precision ( $\sim 15$  digits), we need  $N \sim 10^{30}$ , which is infeasible by direct summation. We address acceleration techniques in Section 3.

**Numerical acceleration.** To circumvent the slow convergence, we use:

- (a) The Borwein–Cohen–Villegas–Zagier acceleration [12]: transforms the conditionally convergent eta series into a geometrically convergent series with error  $O(3^{-N})$ , requiring only  $N \sim 40$  terms for double precision.
- (b) The Euler–Maclaurin summation formula applied to the non-alternating sum  $S_N(s)$ , combined with the identity  $\eta(s) = (1 - 2^{1-s})\zeta(s)$  and the Riemann–Siegel formula for  $\zeta$ .

Both methods are detailed in Section 3.

*Remark 2.3* (Trivial zeros). The trivial zeros of  $\zeta(s)$  at  $s = -2, -4, -6, \dots$  lie outside the critical strip (they have  $\text{Re}(s) < 0$ ). Since the eta series only converges for  $\text{Re}(s) > 0$ , and we restrict our analysis to  $0 < \beta < 1$ , the trivial zeros do not appear in our framework. Furthermore, the prefactor  $(1 - 2^{1-s})$  does not vanish at the trivial zeros (since  $1 - s = 3, 5, 7, \dots$  gives  $2^{1-s} = 2^3, 2^5, \dots \neq 1$ ).

**Why zeros of  $\mathcal{L}$  correspond to DQPTs.** A DQPT is defined as a non-analyticity (cusp, discontinuity, or divergence) in the rate function  $\mathcal{F}(\beta, t)$ , not merely a zero of the order parameter  $\mathcal{L}(\beta, t)$ . We show these are equivalent in this framework.

Consider the rate function  $\mathcal{F}_1(\beta, t) = -(1/\ln N) \ln |\mathcal{L}_N(\beta, t)|$  as  $N \rightarrow \infty$ :

- (i) For  $(\beta, t)$  such that  $\eta(\beta + it) \neq 0$ : the numerator  $\eta_N(s) \rightarrow \eta(s) \neq 0$ , a bounded nonzero analytic function of  $s$ . Write  $\eta_N(s) = \eta(s) + O(N^{-\beta})$  by the Dirichlet test remainder bound. The denominator  $Z_N(\beta) = N^{1-\beta}/(1-\beta) + O(1)$  by Euler–Maclaurin [11]. Hence

$$|\mathcal{L}_N| = |\eta(s)| (1-\beta) N^{\beta-1} (1 + O(N^{-\min(\beta, 1-\beta)})). \quad (2.29)$$

The leading behavior gives  $\mathcal{F}_1 \rightarrow (1-\beta)$ , a smooth function.

- (ii) At a zero  $s_0 = \beta_0 + it_0$  of  $\eta$  (equivalently, of  $\zeta$ ): write  $\eta(s) = \eta'(s_0)(s - s_0) + O(|s - s_0|^2)$  near  $s_0$  (assuming simple zero<sup>4</sup>). At  $s = s_0$  exactly:  $\eta_N(s_0) = O(N^{-\beta_0})$  by the Dirichlet test remainder bound (since the full series converges to  $\eta(s_0) = 0$ ). Hence  $|\mathcal{L}_N(\beta_0, t_0)| = O(N^{\beta-\beta_0-1})$  and  $\mathcal{F}_1 \rightarrow (\beta_0 + 1 - \beta)$ .

The rate function jumps discontinuously from  $(1-\beta)$  to  $(\beta_0 + 1 - \beta)$  at the zero, a discontinuity of magnitude  $\beta_0$ . Since  $\beta_0 > 0$  for all nontrivial zeros, this is a genuine non-analyticity—a first-order DQPT in the classification of Heyl et al. [7].

More precisely, the derivative  $d\mathcal{F}_1/dt$  diverges at  $t = t_0$  (the rate function has a cusp), because  $|\mathcal{L}_N|$  passes through a minimum of order  $N^{\beta-\beta_0-1}$  and the logarithm amplifies this into a peak of height  $O(\ln N)$  in  $\mathcal{F}_1$ . In the thermodynamic limit, this becomes a genuine singularity.

Therefore: zeros of  $\mathcal{L}$  correspond one-to-one with DQPTs (singularities of  $\mathcal{F}_1$ ). This completes the heuristic argument.

*Remark 2.4 (Rigor).* The above argument is carried out at the level of formal asymptotics. A fully rigorous statement requires establishing (i) the existence of the thermodynamic limit  $N \rightarrow \infty$  for  $\mathcal{F}_1$ , with uniform convergence in  $(\beta, t)$  on compact subsets of  $(0, 1) \times \mathbb{R}$  away from zeros, and (ii) that the singularity structure persists in the limit. We verify both conditions numerically (Section 4.4).

*Remark 2.5 (Lehmer phenomenon).* Occasionally, two zeta zeros lie extremely close together. A notable early example is the pair near  $t \approx 7005$  (gap approximately 0.04), identified by Lehmer [13]. Much smaller gaps ( $< 10^{-3}$ ) are known at larger heights [3]. Our zero-detection algorithm (Section 3.3) accounts for this with adaptive  $t$ -refinement.

<sup>4</sup>For zeros of multiplicity  $m$ , see Remark 2.6.

**Catastrophic cancellation at large  $t$ .** For large  $t$ , the summand  $n^{-(\beta+it)} = n^{-\beta} e^{-it \ln n}$  involves rapidly oscillating phases. The partial sum  $S_N(s) = \sum_{n=1}^N n^{-s}$  exhibits extensive cancellation. By the variance heuristic (treating phases as approximately random), the expected magnitude is  $O((\sum_{n=1}^N n^{-2\beta})^{1/2})$ . For  $\beta = \frac{1}{2}$ , this gives  $O(\sqrt{\ln N})$ ; for  $\beta > \frac{1}{2}$ ,  $O(1)$ ; for  $\beta < \frac{1}{2}$ ,  $O(N^{1/2-\beta})$ . More rigorous bounds follow from van der Corput estimates [11]:  $|S_N(\beta + it)| = O(N^{1-\beta} t^{-1/2} + t^{1/2})$  for  $t > 0$ .

The practical consequence: with  $N$  terms each contributing  $O(\varepsilon_{\text{mach}})$  rounding error, the accumulated absolute error is  $O(N\varepsilon_{\text{mach}})$  without compensation (or  $O(\varepsilon_{\text{mach}})$  with Kahan summation [14]). We switch to the Riemann–Siegel formula for  $t > 10^3$  (Section 3.2.1).

*Remark 2.6* (Multiplicity and Robustness of Detection). If a zeta zero  $\rho_0$  has multiplicity  $m > 1$  (widely conjectured not to occur, but not proven), two effects must be distinguished:

- (i) The *infinite series*  $\eta(s)$  vanishes to order  $m$  at  $\rho_0$ :  $\eta(s) = c_m(s - \rho_0)^m + O(|s - \rho_0|^{m+1})$ . This determines the *width* of the dip in  $|\eta(s)|$  around  $\rho_0$ .
- (ii) The *partial sum*  $\eta_N(\rho_0) = O(N^{-\beta_0})$  by the Dirichlet test, *independently* of  $m$ .

Consequently, the rate function jump is  $\beta_0$ , independent of  $m$ . However, zeros of even multiplicity do not produce a sign change in  $Z_H(t)$ . To ensure a scientifically rigorous search that is robust against such cases (or GUE violations), our algorithm monitors local minima of the magnitude  $|\mathcal{L}(\beta, t)|$  in addition to sign changes of  $\mathcal{G}(\beta, t)$ .

### 2.2.2 System 2: Generalized Loschmidt Amplitude

The generalized Loschmidt amplitude is defined as

$$\mathcal{G}(\beta, t) = \frac{1}{2Z(\beta)} \left[ e^{i\vartheta(t)} S_N(\beta + it) + e^{-i\vartheta(t)} S_N(\beta - it) \right], \quad (2.30)$$

where  $\vartheta(t)$  is the Riemann–Siegel theta function:

$$\vartheta(t) = \text{Im}(\ln \Gamma(\tfrac{1}{4} + \tfrac{it}{2})) - \tfrac{t}{2} \ln \pi. \quad (2.31)$$

Since  $S_N(\beta - it) = \overline{S_N(\beta + it)}$  for real  $\beta$ , we have

$$\mathcal{G}(\beta, t) = \frac{1}{Z(\beta)} \text{Re}[e^{i\vartheta(t)} S_N(\beta + it)]. \quad (2.32)$$

While  $Z_H(t)$  is exactly real in the limit  $N \rightarrow \infty$  due to the functional equation, for finite  $N$ , the partial sum  $S_N$  does not satisfy the  $\zeta(s) \leftrightarrow \zeta(1-s)$  symmetry.

Consequently,  $\mathcal{G}_N(\beta, t)$  is only an approximation whose deviations from the real axis (or the target function) scale as the tail of the Dirichlet series, typically  $O(N^{-\beta})$ . At  $\beta = 1/2$ , this implies a remainder of  $O(N^{-1/2})$ , which must be accounted for in the validation tolerances.

**Divergence of  $Z(\beta)$  in System 2.** The same divergence issue arises as in System 1. The resolution is identical: we work with the finite- $N$  quantity  $\mathcal{G}_N(\beta, t)$  and extract the rate function  $\mathcal{F}_2(\beta, t) = -\lim_{N \rightarrow \infty} (1/\ln N) \ln |\mathcal{G}_N(\beta, t)|$ , which has the same behavior as  $\mathcal{F}_1$ : singularities at zeta zeros, smooth elsewhere.

**Connection to the Hardy  $Z$ -function.** At  $\beta = \frac{1}{2}$ , in the limit  $N \rightarrow \infty$ , the partial sum  $S_N(\frac{1}{2} + it)$  converges to  $\zeta(\frac{1}{2} + it)$ . The Riemann–Siegel theta function is chosen precisely so that

$$Z_H(t) = e^{i\vartheta(t)} \zeta(\tfrac{1}{2} + it) \quad (2.33)$$

is real-valued for all real  $t$  [15]. This is the Hardy  $Z$ -function. Its sign changes correspond one-to-one with simple zeros of  $\zeta(\frac{1}{2} + it)$  on the critical line.

**Why zeros of  $\mathcal{G}$  correspond to zeta zeros.** As with System 1,  $Z_N(\frac{1}{2})$  diverges as  $N \rightarrow \infty$ , so  $\mathcal{G}_N(\frac{1}{2}, t)$  itself tends to zero. The physically meaningful statement concerns the *ratio structure*: the numerator  $\text{Re}[e^{i\vartheta(t)} S_N(\frac{1}{2} + it)]$  converges to  $Z_H(t)$  independently of the divergent denominator. The zeros of  $\mathcal{G}_N$  (detected at finite  $N$ ) correspond to the zeros of  $Z_H(t)$  in the limit. Since  $Z_H(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$  (because  $e^{i\vartheta(t)} \neq 0$ ), and Hardy [15] proved infinitely many zeros lie on the critical line, System 2 detects exactly the nontrivial zeros on the critical line.

*Remark 2.7* (System 2 false positives at  $\beta \neq \frac{1}{2}$ ). For  $\beta \neq \frac{1}{2}$ ,  $\mathcal{G}(\beta, t)$  can vanish even when  $\zeta(\beta + it) \neq 0$ : since  $e^{i\vartheta(t)} \zeta(\beta + it)$  is generally complex for  $\beta \neq \frac{1}{2}$ , its real part vanishes whenever the phase of  $e^{i\vartheta(t)} \zeta(\beta + it)$  equals  $\pm\pi/2$ . These are *phase-alignment zeros*, not zeta zeros, and constitute false positives.

*Consequence for experimental design:* System 2 is reliable for zero detection *only* at  $\beta = \frac{1}{2}$ . For off-critical-line scanning ( $\beta \neq \frac{1}{2}$ ), System 1 (the  $\mathcal{L}$  function, based on the eta series) must be used. The experimental design (Section 4.2) is structured accordingly.

## 2.3 The RH–DQPT Equivalence

**Theorem 2.8** (RH–DQPT Equivalence). *The Riemann Hypothesis is equivalent to the following statement: for both System 1 and System 2, dynamical quantum phase*

transitions (zeros of the order parameter / non-analyticities of the rate function) occur only at inverse temperature  $\beta = \frac{1}{2}$ .

More precisely:

- (a) RH holds if and only if, for all  $t \in \mathbb{R}$  and all  $\beta \in (0, 1)$  with  $\beta \neq \frac{1}{2}$ :  $\mathcal{L}(\beta, t) \neq 0$ .
- (b) At  $\beta = \frac{1}{2}$  specifically: RH holds if and only if the zeros of  $\mathcal{G}(\frac{1}{2}, t)$  (as a function of  $t$ ) account for all nontrivial zeros of  $\zeta$ . (System 2 cannot be used for off-line scanning due to false positives; statement (a) via System 1 is the operative equivalence.)

*Proof. Part (a).* ( $\Rightarrow$ ) If RH holds, all nontrivial zeros have  $\text{Re}(s) = \frac{1}{2}$ . Then  $\eta(\beta + it) = (1 - 2^{1-(\beta+it)})\zeta(\beta + it) \neq 0$  for  $\beta \neq \frac{1}{2}$  in  $(0, 1)$  (since  $\zeta$  has no zeros there and the prefactor is nonzero for  $0 < \beta < 1$  as shown in Step 3). Hence  $\mathcal{L}(\beta, t) \neq 0$ .

( $\Leftarrow$ ) If RH fails, there exists  $\zeta(\beta_0 + it_0) = 0$  with  $\beta_0 \neq \frac{1}{2}$  and  $0 < \beta_0 < 1$ . Then  $\eta(\beta_0 + it_0) = 0$ , so  $\mathcal{L}(\beta_0, t_0) = 0$ , giving a DQPT at  $\beta_0 \neq \frac{1}{2}$ . Contrapositive: no DQPTs off the critical line implies RH.  $\square$

**What it would mean to find a DQPT at  $\beta \neq \frac{1}{2}$ .** This would constitute a computational disproof of the Riemann Hypothesis: the existence of a nontrivial zero of  $\zeta(s)$  with  $\text{Re}(s) \neq \frac{1}{2}$ , with immediate consequences for analytic number theory and the distribution of primes.

## 2.4 Free Energy and Singularities

The rate function (free energy density) for System 1 is  $\mathcal{F}_1(\beta, t) = -\lim_{N \rightarrow \infty} (1/\ln N) \ln |\mathcal{L}_N(\beta, t)|$  as derived in Section 2.2.1, Step 5. The values are:

$$\mathcal{F}_1(\beta, t) = 1 - \beta \quad (\text{away from zeros}), \quad (2.34)$$

$$\mathcal{F}_1(\beta, t_0) = \beta_0 + 1 - \beta \quad (\text{at a zero with } \text{Re}(s_0) = \beta_0), \quad (2.35)$$

$$\Delta\mathcal{F}_1 = \beta_0 \quad (\text{DQPT jump magnitude}). \quad (2.36)$$

At  $\beta = \beta_0 = \frac{1}{2}$ :  $\mathcal{F}_1 = 1$  at the zero,  $\mathcal{F}_1 = \frac{1}{2}$  away. The DQPT jump is  $\frac{1}{2}$ .

*Remark 2.9* (Normalization conventions). Wei et al. [1] use a different normalization employing  $\log_2$  rather than  $\ln$ . Under their convention,  $\mathcal{F}_1^{\text{Wei}} = \ln 2 \cdot \mathcal{F}_1^{\text{nat}}$ , giving  $\mathcal{F}_1^{\text{Wei}} = \ln 2$  at zeros and  $(1 - \beta) \ln 2$  away. We use natural logarithms throughout; numerical values differ by a factor of  $\ln 2$ .

### 3 Numerical Methods

#### 3.1 Computation of Partial Dirichlet Series

The fundamental computational task is evaluating the partial Dirichlet series:

$$S_N(s) = \sum_{n=1}^N n^{-s}, \quad s = \beta + it. \quad (3.1)$$

**Direct computation.** Each term is computed as

$$n^{-s} = n^{-\beta} n^{-it} = n^{-\beta} e^{-it \ln n}, \quad (3.2)$$

and separated into real and imaginary parts:

$$\operatorname{Re}(n^{-s}) = n^{-\beta} \cos(t \ln n), \quad (3.3)$$

$$\operatorname{Im}(n^{-s}) = -n^{-\beta} \sin(t \ln n). \quad (3.4)$$

This decomposition avoids complex exponentiation and uses only real arithmetic (two multiplications, one cosine, one sine per term), which is more numerically stable and cache-friendly.

**Alternating series.** For System 1, we need the alternating partial sum

$$\eta_N(s) = \sum_{n=1}^N (-1)^{n+1} n^{-s}, \quad (3.5)$$

obtained by multiplying each term by  $(-1)^{n+1} = +1$  for odd  $n$ ,  $-1$  for even  $n$ .

**Numerical stability.** The terms  $n^{-\beta}$  decrease monotonically for  $\beta > 0$ , ensuring numerical stability of the running sum (no catastrophic cancellation in the absolute sense). However, for the alternating series with small  $\beta$ , consecutive terms nearly cancel, leading to loss of relative precision. For  $\beta = 0.5$  and large  $n$ , consecutive terms differ by  $O(1/n)$ , so the partial sum has relative precision degraded by a factor of roughly  $\sqrt{N}$ . For  $N = 10^5$ , this costs approximately 2–3 decimal digits, which is acceptable in double precision ( $\approx 15$ – $16$  significant digits).

**Precomputation.** For a fixed  $\beta$ , the values  $n^{-\beta}$  ( $n = 1, \dots, N$ ) are independent of  $t$  and can be precomputed and stored. This reduces the per-point cost to  $O(N)$  multiplications and trigonometric evaluations.

**Error bound.** The truncation error for the ordinary Dirichlet series is

$$|S_N(s) - \zeta(s)| \leq \sum_{n=N+1}^{\infty} n^{-\beta}. \quad (3.6)$$

By integral comparison:

$$\sum_{n=N+1}^{\infty} n^{-\beta} \leq \int_N^{\infty} x^{-\beta} dx = \frac{N^{1-\beta}}{\beta-1}, \quad \beta > 1. \quad (3.7)$$

For  $0 < \beta \leq 1$  (the critical strip), the ordinary Dirichlet series does not converge, but the alternating series converges. The truncation error for the alternating series is bounded by the Dirichlet test (monotone coefficients, bounded partial sums of  $(-1)^{n+1}$ ):

$$|\eta_N(s) - \eta(s)| \leq C(s) (N+1)^{-\beta}, \quad (3.8)$$

where  $C(s) = O(1)$  for bounded  $|s|$ . For  $\beta = 1/2$  and  $N = 10^4$ : error  $\lesssim 10^{-2}$ ; for  $N = 10^6$ : error  $\lesssim 10^{-3}$ . This is the fundamental limitation of direct summation in the critical strip.

**Euler–Maclaurin acceleration.** For improved convergence, we apply the Euler–Maclaurin summation formula to the tail:

$$\begin{aligned} \sum_{n=N+1}^M n^{-s} &\approx \int_N^M x^{-s} dx + \frac{1}{2}(N^{-s} + M^{-s}) \\ &+ \sum_{k=1}^p \frac{B_{2k}}{(2k)!} s(s+1) \cdots (s+2k-2) (N^{-(s+2k-1)} - M^{-(s+2k-1)}) + R_p, \end{aligned} \quad (3.9)$$

where  $B_{2k}$  are Bernoulli numbers and  $R_p$  is the remainder. Taking  $M \rightarrow \infty$  and using  $p$  terms of the asymptotic expansion, the error bound must account for the rising factorial  $|s(s+1) \cdots (s+2p-2)|$ , which grows as  $|s|^{2p-1} \sim t^{2p-1}$  for large  $t$ . The correct error estimate is

$$|R_p| = O(|s|^{2p-1} N^{-(\beta+2p-1)}). \quad (3.10)$$

This means that for large  $t$ , the Euler–Maclaurin correction terms grow and the asymptotic expansion eventually diverges (as is typical of asymptotic series). The optimal number of correction terms  $p$  depends on both  $N$  and  $t$ .

**Precision table for Euler–Maclaurin correction ( $\beta = 1/2$ ).** In the following table,  $|s|^{2p-1}$  denotes the magnitude of the rising factorial  $s(s+1) \cdots (s+2p-2)$ , approximated as  $|s|^{2p-1}$  for  $|s| \gg 1$ . With  $s = 1/2 + it$ , we have  $|s| \approx t$  for large  $t$ .

**Table 1:** Euler–Maclaurin effective error at  $\beta = 1/2$ .

| $N$    | $t$    | $p$ | $ s ^{2p-1}$ | $N^{-(\beta+2p-1)}$ | Effective error   |
|--------|--------|-----|--------------|---------------------|-------------------|
| $10^4$ | $10^2$ | 3   | $10^{10}$    | $10^{-22}$          | $\sim 10^{-12}$   |
| $10^4$ | $10^2$ | 5   | $10^{18}$    | $10^{-38}$          | $\sim 10^{-20}$   |
| $10^4$ | $10^4$ | 3   | $10^{20}$    | $10^{-22}$          | $\sim 10^{-2}$    |
| $10^4$ | $10^4$ | 5   | $10^{36}$    | $10^{-38}$          | $\sim 10^{-2}$    |
| $10^5$ | $10^2$ | 5   | $10^{18}$    | $10^{-47.5}$        | $\sim 10^{-29.5}$ |
| $10^3$ | $10^2$ | 3   | $10^{10}$    | $10^{-16.5}$        | $\sim 10^{-6.5}$  |

The table reveals that for large  $t$ , the Euler–Maclaurin correction becomes unreliable: at  $t = 10^4$  with any  $p$ , the rising factorial overwhelms the  $N^{-(\beta+2p-1)}$  decay. This is the fundamental reason why the Riemann–Siegel formula (Section 3.2.1) is essential for  $t > 10^3$ . For moderate  $t$  ( $t < 10^3$ ), Euler–Maclaurin with  $p = 5$  and  $N = 10^4$  provides  $\sim 20$  digits of precision, far exceeding double-precision limits.

*Remark 3.1* (On the relationship between acceleration and the DQPT framework). A methodological subtlety deserves comment. The DQPT framework is formulated in terms of the thermodynamic limit  $N \rightarrow \infty$  of the *finite* partial sums  $\mathcal{L}_N(\beta, t)$ , and the physics resides in the finite- $N$  scaling behavior (Section 2.2.1). When we apply Euler–Maclaurin summation or Borwein acceleration to compute the “true” value  $\eta(s)$  (the infinite-sum limit), we are effectively bypassing the finite- $N$  physics and returning to standard analytic number theory. This does not invalidate the computation, but it means that the accelerated results should be understood as *high-precision evaluations of the same mathematical function*, not as simulations of a physical system at finite  $N$ . We use both approaches: unaccelerated partial sums (to verify finite- $N$  scaling predictions of the DQPT theory) and accelerated evaluations (to achieve the precision needed for reliable zero detection and comparison with published tables).

**Borwein–Cohen–Villegas–Zagier acceleration.** For the eta function specifically, Borwein [12] showed that the following identity provides geometric convergence:

$$\eta(s) = -\frac{1}{d_n} \sum_{k=0}^{n-1} (-1)^k (d_k - d_n) (k+1)^{-s}, \quad (3.11)$$

where the coefficients  $d_k$  are defined by

$$d_k = n \sum_{j=0}^k \frac{(n+j-1)! 4^j}{(n-j)! (2j)!} \quad (3.12)$$

and  $d_n = d_k|_{k=n}$ .

**Numerical stability of the coefficients.** For  $n = 40$ , the coefficients  $d_k$  grow extremely rapidly ( $d_{40} \sim 10^{23}$ ), exceeding the range of IEEE 754 double precision (which represents integers exactly only up to  $2^{53} \approx 9 \times 10^{15}$ ). To avoid overflow and catastrophic cancellation, we compute the *ratios*  $c_k = (d_k - d_n)/d_n$  directly via a stable recurrence. Define  $e_k = d_k/d_n$ ; then  $c_k = e_k - 1$ , and the Borwein formula becomes

$$\eta(s) = - \sum_{k=0}^{n-1} (-1)^k c_k (k+1)^{-s}. \quad (3.13)$$

The ratios  $e_k$  can be computed via the recurrence relation for the  $d_k$  sequence, working entirely in double precision without overflow (since  $0 \leq e_k \leq 1$  for all  $k < n$ ).

**Error bound and region of validity.** The approximation error for the eta function satisfies [12]

$$|\eta(s) - \text{approximation}| \leq 3(3 + \sqrt{8})^{-n} \quad (3.14)$$

for  $s$  in the region  $\text{Re}(s) > 0$ . With  $n = 40$ :

$$|\text{error}_\eta| < 3 \times 5.828^{-40} \approx 10^{-30}. \quad (3.15)$$

If one needs  $\zeta(s)$  instead, it is recovered via  $\zeta(s) = \eta(s)/(1 - 2^{1-s})$ . The resulting error in  $\zeta$  is amplified by the factor  $1/|1 - 2^{1-s}|$ , which diverges near  $s = 1 + 2\pi ik/\log 2$  for  $k \in \mathbb{Z}$ . For computations in the critical strip ( $0 < \text{Re}(s) < 1$ ),  $|1 - 2^{1-s}|$  is bounded away from zero. To see this:  $2^{1-s} = 1$  requires both  $|2^{1-s}| = 2^{\text{Re}(1-s)} = 1$  (i.e.,  $\text{Re}(1-s) = 0$ , hence  $\text{Re}(s) = 1$ ) and  $\arg(2^{1-s}) = \text{Im}(1-s) \log 2 \in 2\pi\mathbb{Z}$ . Since  $\text{Re}(s) = 1$  lies outside the open critical strip  $0 < \text{Re}(s) < 1$ , the prefactor  $1 - 2^{1-s}$  never vanishes inside the strip. Moreover, for  $\text{Re}(s) \in [0.1, 0.9]$ ,  $|1 - 2^{1-s}| \geq c > 0$  for some explicit constant  $c$  (computable for each  $\beta$ ), so the error amplification is bounded.

This geometric convergence rate of  $O(5.828^{-n})$  means  $n = 40$  terms suffice for  $\sim 30$  digits of accuracy—far exceeding double precision requirements. This is our primary method for computing  $\eta(s)$  in the critical strip.

**Kahan compensated summation.** All summation loops use the Kahan compensated summation algorithm [14] to reduce the accumulated floating-point rounding error from  $O(N \varepsilon_{\text{mach}})$  to  $O(\varepsilon_{\text{mach}})$ , independent of  $N$ . The algorithm maintains a running compensation variable that tracks the low-order bits lost in each addition:

**Listing 1:** Kahan compensated summation.

```
double sum = 0.0, c = 0.0;
for (int i = 0; i < N; ++i) {
```

```

double y = x[i] - c;
double t = sum + y;
c = (t - sum) - y;
sum = t;
}

```

*Remark 3.2* (FMA and GPU execution). CUDA compiles with fused multiply-add (FMA) instructions enabled by default. FMA computes  $a \cdot b + c$  with a single rounding (rather than the two roundings of separate multiply and add), which changes the numerical result at the level of  $O(\varepsilon)$  per operation. This has two consequences:

1. The Kahan summation algorithm relies on the rounding behavior of individual additions. With FMA enabled, the compiler may fuse the compensation steps, potentially defeating the purpose of the algorithm. To ensure correct Kahan summation on GPU, use `__dadd_rn()` and `__dsub_rn()` intrinsics (round-to-nearest double-precision add/subtract) or compile with `--fmad=false` for the relevant kernels.
2. CPU and GPU results will differ at  $O(\varepsilon)$  per operation due to different FMA behavior (CPU code compiled with `-ffp-contract=off` does not use FMA). When comparing CPU and GPU results for validation, the expected discrepancy is  $O(N\varepsilon)$  without Kahan summation, or  $O(\varepsilon)$  with correct Kahan summation on both platforms.

### 3.2 Riemann–Siegel Theta Function

The Riemann–Siegel theta function is defined by

$$\vartheta(t) = \operatorname{Im}(\log \Gamma(\tfrac{1}{4} + \tfrac{it}{2})) - \frac{t}{2} \log \pi. \quad (3.16)$$

**Stirling asymptotic expansion.** For  $|t| \gg 1$ , we use the asymptotic expansion of  $\log \Gamma$  via Stirling’s series. Setting  $z = 1/4 + it/2$ :

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + \sum_{k=1}^p \frac{B_{2k}}{2k(2k-1) z^{2k-1}} + R_p(z). \quad (3.17)$$

Taking the imaginary part and substituting  $z = 1/4 + it/2$ , we obtain [10]<sup>5</sup>

$$\vartheta(t) = \frac{t}{2} \ln \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + \frac{7}{5760t^3} + \frac{31}{80640t^5} - \frac{127}{430080t^7} + O(t^{-9}). \quad (3.18)$$

For our implementation, we use the first four correction terms (through  $t^{-7}$ ), which provides more than adequate precision for  $t \geq 10$ .

<sup>5</sup>The first two correction terms ( $1/(48t)$  and  $7/(5760t^3)$ ) are well known; the higher-order coefficients follow from the Stirling series and are tabulated in Edwards [10], Section 6.5.

**Error bound.** The error after  $p$  terms of the Stirling expansion is bounded by

$$|R_p| \leq \frac{|B_{2p+2}|}{(2p+2)(2p+1)|z|^{2p+1}}. \quad (3.19)$$

For  $t > 10$  and  $p = 4$  (using terms through  $t^{-7}$ ):  $|R_4| < 10^{-12}$ , which is adequate for double precision. For  $t > 50$  and  $p = 3$  (through  $t^{-5}$ ):  $|R_3| < 10^{-14}$ .

**Validity range.** The asymptotic expansion is valid for  $t > 0$  but achieves useful precision only for  $t > t_{\min}$ . We require:

- For double precision ( $10^{-15}$ ):  $t_{\min} \approx 10$  (with 4 correction terms).
- For extended precision ( $10^{-30}$ ):  $t_{\min} \approx 5$  (with 8 correction terms).

For  $t < t_{\min}$ , we fall back to numerical evaluation of  $\log \Gamma$  using the Lanczos approximation or a rational Chebyshev approximation for  $\Gamma$  in the relevant region.

*Remark 3.3* (Branch cuts and small  $t$ ). The function  $\vartheta(t) = \text{Im}(\log \Gamma(1/4 + it/2)) - (t/2) \log \pi$  involves the complex logarithm of the Gamma function, which has branch cuts. For small  $t$  ( $t < 10$ ), the argument  $1/4 + it/2$  is close to the real axis where  $\Gamma$  has poles at  $0, -1, -2, \dots$ , and the principal branch of  $\log \Gamma$  requires careful handling to avoid discontinuities. Our implementation strategy:

- For  $t \geq 10$ : use the Stirling asymptotic expansion (4+ correction terms), which is well away from branch cut issues and provides adequate precision.
- For  $t < 10$ : use the Lanczos approximation for  $\Gamma(z)$  (with reflection formula for  $\text{Re}(z) < 0.5$ ), or equivalently use the recurrence  $\Gamma(z+1) = z \Gamma(z)$  to shift the argument into a region where the Lanczos series converges rapidly. This avoids the branch cut issues of  $\log \Gamma$  by computing  $\Gamma$  directly and then taking the logarithm.

In practice, the first zeta zero is at  $t \approx 14.13$ , so the small- $t$  regime is only relevant for theta function evaluation at sub-zero-height positions (used in Gram point computation and the Riemann–von Mangoldt formula).

### 3.2.1 The Riemann–Siegel Formula

The Riemann–Siegel (RS) formula provides an efficient method for computing the Hardy  $Z$ -function  $Z_H(t) = e^{i\vartheta(t)} \zeta(1/2 + it)$  directly, requiring only  $O(\sqrt{t/(2\pi)})$  terms rather than  $O(t)$  terms for direct summation [11, 16]. This is the standard method for large-scale zero computations.

**Explicit formula.** The RS formula is

$$Z_H(t) = 2 \sum_{n=1}^M n^{-1/2} \cos(\vartheta(t) - t \ln n) + R(t), \quad (3.20)$$

where  $M = \lfloor \sqrt{t/(2\pi)} \rfloor$  is the main sum cutoff and  $R(t)$  is the remainder term.

**First correction term.** The leading-order remainder is given by the Riemann–Siegel correction [16]:

$$R_0(t) = (-1)^{M-1} \left( \frac{t}{2\pi} \right)^{-1/4} \Psi(p), \quad (3.21)$$

where  $p = \sqrt{t/(2\pi)} - M$  is the fractional part and  $\Psi(p)$  is the Riemann–Siegel function:

$$\Psi(p) = \frac{\cos(2\pi(p^2 - p - 1/16))}{\cos(2\pi p)}. \quad (3.22)$$

Higher-order correction terms  $R_1, R_2, \dots$  involve derivatives of  $\Psi$  and powers of  $(t/(2\pi))^{-1/2}$ , each successive term reducing the error by an additional factor of  $O(t^{-1/2})$ :

$$R(t) = R_0(t) + R_1(t) + R_2(t) + \dots, \quad (3.23)$$

where  $|R_k(t)| = O(t^{-(2k+3)/4})$  (Siegel, 1932; Gabcke [16]).

**Error bounds after successive corrections.** The following bounds are verified against Titchmarsh [11] and Gabcke [16]:

$$\text{Main sum only:} \quad |Z_H(t) - \text{main sum}| = O(t^{-3/4}), \quad (3.24)$$

$$\text{Main sum} + R_0 : \quad |\text{error}| = O(t^{-5/4}), \quad (3.25)$$

$$\text{Main sum} + R_0 + R_1 : \quad |\text{error}| = O(t^{-7/4}). \quad (3.26)$$

Quantitatively (Gabcke [16]): the error of the main sum alone is less than  $(3/2)t^{-3/4}$  for  $t > 125$ . Including the  $R_0$  correction, the error is less than  $0.053t^{-5/4}$  for  $t \geq 200$ .

For  $t = 10^4$ : the uncorrected remainder is  $O(t^{-3/4}) = O(10^{-3})$ . Including  $R_0$ , the error improves to  $O(t^{-5/4}) = O(10^{-5})$ . Including  $R_0 + R_1$ , the error is  $O(t^{-7/4}) = O(10^{-7})$ .

**Computational advantage.** The main sum requires only  $M = \lfloor \sqrt{t/(2\pi)} \rfloor$  terms:

| $t$       | $M$          |
|-----------|--------------|
| $10^4$    | $\sim 40$    |
| $10^6$    | $\sim 399$   |
| $10^8$    | $\sim 3989$  |
| $10^{10}$ | $\sim 39894$ |

Compare this to direct summation of the Dirichlet series, which requires  $N \gg t$  for adequate convergence. The RS formula is essential for computations at large height.

**Implementation notes.** Each term in the main sum requires evaluation of  $n^{-1/2} \cos(\vartheta(t) - t \ln n)$ . Since  $n^{-1/2}$  is independent of  $t$ , it can be precomputed. The argument  $\vartheta(t) - t \ln n$  must be reduced modulo  $2\pi$  carefully to avoid loss of precision for large  $t$  (see Section 5.2 on trigonometric evaluation).

### 3.3 Zero Detection

#### 3.3.1 Bisection on the Hardy $Z$ -function

The Hardy  $Z$ -function  $Z_H(t)$  is real-valued, and its sign changes correspond to simple zeros of  $\zeta(1/2 + it)$ . Our primary zero-detection method is:

1. Evaluate  $Z_H(t)$  on a uniform grid  $t_0, t_0 + \Delta t, t_0 + 2\Delta t, \dots$
2. Identify intervals  $[t_k, t_{k+1}]$  where  $Z_H$  changes sign.
3. Refine each sign change by bisection to precision  $\varepsilon$ .

**Grid spacing.** The average spacing between consecutive zeros near height  $T$  is

$$\delta_{\text{zero}} \sim \frac{2\pi}{\log(T/(2\pi))}. \quad (3.27)$$

For  $T = 100$ :  $\delta_{\text{zero}} \approx 2.3$ . For  $T = 1000$ :  $\delta_{\text{zero}} \approx 1.2$ . For  $T = 10000$ :  $\delta_{\text{zero}} \approx 0.86$ . We choose  $\Delta t = \delta_{\text{zero}}/4$  to ensure we do not miss any zeros (assuming zeros are roughly uniformly spaced, which is a reasonable heuristic for the initial scan).

#### 3.3.2 Gram's Law and Its Failures

Gram points are the values  $g_n$  defined by  $\vartheta(g_n) = n\pi$  for  $n = 0, 1, 2, \dots$ . Gram's "law" (actually a heuristic) states that  $(-1)^n Z_H(g_n) > 0$ , i.e.,  $Z_H$  changes sign between consecutive Gram points. This holds for approximately 73% of Gram intervals [17].

We use Gram points as an initial grid but do not rely on Gram's law for correctness. Instead, we use finer subdivision wherever Gram's law fails (i.e., where  $Z_H$  has the same sign at two consecutive Gram points).

#### 3.3.3 Turing's Method for Zero Count Certification

Turing [18] developed a method to certify that all zeros up to height  $T$  have been found. The method relies on the Riemann–von Mangoldt formula for the zero-counting function:

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + S(T) + O\left(\frac{1}{T}\right), \quad (3.28)$$

where  $S(T) = (1/\pi) \arg \zeta(1/2 + iT)$  satisfies  $|S(T)| = O(\log T)$  unconditionally<sup>6</sup>. If the number of detected sign changes of  $Z_H$  in  $[0, T]$  agrees with the nearest integer to  $(T/(2\pi)) \log(T/(2\pi)) - T/(2\pi) + 7/8$ , then all zeros in that range have been found (up to the uncertainty introduced by  $S(T)$ ).

We implement this check as a post-hoc validation: after detecting zeros up to height  $T$ , we verify that the count matches the Riemann–von Mangoldt prediction.

### 3.3.4 Expected Zero Density

The number of nontrivial zeros with  $0 < \text{Im}(s) \leq T$  is given by the Riemann–von Mangoldt formula (3.28). The leading terms give the approximation:

$$N(T) \approx \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8}. \quad (3.29)$$

This gives:

| $T$     | $N(T)$         |
|---------|----------------|
| 100     | $\sim 29$      |
| 1,000   | $\sim 649$     |
| 10,000  | $\sim 10,142$  |
| 100,000 | $\sim 138,069$ |

## 3.4 Convergence Analysis

### 3.4.1 Convergence of $L(\beta, t)$ with $N$

For fixed  $s = \beta + it$  with  $0 < \beta < 1$ , the alternating partial sum  $\eta_N(s)$  converges to  $\eta(s)$  as  $N \rightarrow \infty$ . The rate of convergence is

$$|\eta_N(s) - \eta(s)| = O(N^{-\beta}). \quad (3.30)$$

This is the dominant source of error. For  $\beta = 1/2$ :

| $N$    | Error          |
|--------|----------------|
| $10^4$ | $\sim 10^{-2}$ |
| $10^6$ | $\sim 10^{-3}$ |
| $10^8$ | $\sim 10^{-4}$ |

This slow convergence is the fundamental computational bottleneck. To achieve precision  $\delta$ , we need  $N \sim \delta^{-1/\beta} = \delta^{-2}$  for  $\beta = 1/2$ . This is much worse than the Riemann–Siegel formula, which achieves precision  $\delta$  with  $O(\sqrt{T})$  terms, but the direct approach has the virtue of simplicity and direct connection to the physical (DQPT) interpretation.

<sup>6</sup>See Trudgian [17] for modern explicit bounds. Note that  $S(T)$  is *not* bounded by a constant; the bound  $|S(T)| = O(\log T)$  reflects the irregular distribution of zeta zeros.

**Improvement via Euler–Maclaurin.** With  $p$  terms of Euler–Maclaurin correction applied to the tail sum beyond  $N$ , the effective error becomes  $O(|s|^{2p-1} N^{-(\beta+2p-1)})$  (see Section 3.1 for the full bound and precision table). For  $p = 5$ ,  $\beta = 1/2$ , and moderate  $t$  ( $t < 10^4$ ): the error is well below double-precision limits for  $N \geq 10^4$ . This makes Euler–Maclaurin-corrected summation practical for moderate  $t$ . For large  $t$ , the Riemann–Siegel formula (Section 3.2.1) is preferable.

### 3.4.2 Required $N$ as a Function of $t$ and Precision

The terms  $n^{-s} = n^{-\beta} e^{-it \ln n}$  oscillate with frequency proportional to  $t$ . The Nyquist criterion requires that we sample these oscillations adequately, but since we are summing (not sampling) the series, the oscillation frequency does not directly increase the required  $N$ . However, near zeros of  $\zeta$ , the partial sums exhibit cancellation at scale  $O(N^{-\beta})$ , so detecting zeros to relative precision  $\delta$  requires

$$N \geq \delta^{-1/\beta}. \quad (3.31)$$

For  $\beta = 1/2$  and  $\delta = 10^{-6}$  (sufficient to distinguish a zero from a near-miss):  $N \geq 10^{12}$ . This is impractical without acceleration techniques.

With Euler–Maclaurin ( $p = 5$ ) and accounting for the  $|s|^{2p-1}$  factor: the required  $N$  satisfies  $|s|^{2p-1} N^{-(\beta+2p-1)} < \delta$ , giving

$$N \geq \left( \frac{|s|^{2p-1}}{\delta} \right)^{1/(\beta+2p-1)}. \quad (3.32)$$

For  $t = 100$ ,  $|s| \approx 100$ ,  $p = 5$ ,  $\beta = 0.5$ :  $N \geq (10^{18}/10^{-6})^{1/9.5} = 10^{24/9.5} \approx 336$ . For  $t = 10^4$ :  $N \geq (10^{36}/10^{-6})^{1/9.5} = 10^{42/9.5} \approx 26,300$ . These are achievable, showing that Euler–Maclaurin correction is essential for practical computation at high precision, though the required  $N$  grows with  $t$ .

### 3.4.3 Computational Complexity

Direct summation:  $O(N)$  operations per  $(\beta, t)$  point. For  $M$  grid points in the  $(\beta, t)$  plane, the total cost is  $O(N \cdot M)$ .

With Euler–Maclaurin:  $O(N + p)$  per point, where  $p$  is small (5–10). For a moderate  $N$  ( $10^3$  to  $10^4$ ), the cost is dominated by the direct sum, giving  $O(N \cdot M)$  total.

For a grid of  $M_\beta = 100$  values of  $\beta$  and  $M_t = 10,000$  values of  $t$ , with  $N = 10^4$ :

$$\text{Total operations} \sim 10^4 \times 10^6 = 10^{10}. \quad (3.33)$$

On a single CPU core, this workload requires substantial time (minutes to hours depending on architecture and instruction set). While GPU parallelism drastically reduces wall-clock time, the hardware-level bottleneck of FP64 transcendental units must be considered (Section 3.5.3).

## 3.5 GPU Parallelization Strategy

### 3.5.1 Parallelism Structure

The computation of  $L(\beta, t)$  or  $G(\beta, t)$  at each  $(\beta, t)$  point is completely independent of all other points. This is an embarrassingly parallel workload, ideal for GPU execution.

#### Parallelization scheme.

- Each CUDA thread computes one  $(\beta, t)$  point.
- The thread iterates over  $n = 1, \dots, N$ , accumulating the partial sum.
- The result (a complex number for  $L$ , a real number for  $G$ ) is written to global memory.

### 3.5.2 Memory Model

For each  $\beta$  value, we precompute the array  $\{n^{-\beta}\}_{n=1}^N$  and store it in global memory (or texture memory for cached access). Each thread reads this array and computes the trigonometric factors  $\cos(t \ln n)$  and  $\sin(t \ln n)$  on the fly.

#### Memory requirements.

- Precomputed  $n^{-\beta}$ :  $N \times \text{sizeof}(\text{double}) = N \times 8$  bytes per  $\beta$  value. For  $N = 10^5$  and 100  $\beta$  values: 80 MB (fits in 6 GB VRAM).
- Precomputed  $\ln(n)$ :  $N \times 8$  bytes = 0.8 MB (shared across all  $\beta$  values).
- Output array:  $M_\beta \times M_t \times \text{sizeof}(\text{double})$  per output quantity. For  $100 \times 10,000 = 10^6$  points: 8 MB.
- Total VRAM usage:  $< 100$  MB, well within 6 GB.

### 3.5.3 Expected Throughput

RTX 2060 specifications:

- 1920 CUDA cores.
- 6 GB GDDR6 VRAM.
- Base clock: 1365 MHz.
- FP64 throughput: approximately 210 GFLOPS (1/32 of FP32).

Since our computation involves double-precision transcendental functions (`cos`, `sin`, `exp`, `log`), the throughput is limited by the special function units (SFU).

*Remark 3.4* (SFU throughput for transcendentals). On NVIDIA GPUs, the SFU throughput for double-precision transcendental functions is approximately 1/4 of the FP64 ALU throughput (and for some transcendentals, up to 1/10 due to multi-pass evaluation). Since our workload is transcendental-heavy (each term requires one `cos` and one `sin` evaluation), the effective throughput must be derated by 4–10× compared to the peak FP64 rate.

### Revised throughput estimates.

- Per-point cost:  $N$  multiplications +  $N$  transcendental evaluations. The transcendentals dominate: effective cost  $\sim 10N$  FLOPS equivalent (accounting for SFU pipeline occupancy).
- For  $N = 10^4$ :  $\sim 10^5$  FLOPS-equivalent per point.
- Effective throughput:  $(210 \times 10^9 / 4) / (10^5) \approx 5 \times 10^5$  points/second (conservative; actual throughput depends on occupancy and memory access patterns).

For a  $180 \times 10,000$  grid ( $1.8 \times 10^6$  points): approximately 3–4 seconds. For a  $180 \times 100,000$  grid ( $1.8 \times 10^7$  points): approximately 30–40 seconds. For  $N = 10^5$  and a  $180 \times 10,000$  grid: approximately 30–40 seconds.

These revised estimates (4–10× slower than naive FP64 peak estimates) still show that GPU-accelerated computation is more than sufficient for our parameter ranges. Even with additional overheads (memory latency, register pressure, thread divergence), runtimes remain under a few minutes for the full sweep.

#### 3.5.4 Batch Size Estimation

We organize the computation into batches to manage VRAM and kernel launch overhead:

- Batch size:  $10^5$  to  $10^6$  ( $\beta, t$ ) points per kernel launch.
- Number of batches: total grid size / batch size.
- For a  $100 \times 100,000$  grid:  $10^7 / 10^6 = 10$  batches.

Each batch completes in  $< 1$  second, so the total wall-clock time (including data transfer) is under 10 seconds for the full sweep.

## 4 Experimental Design

### 4.1 Hypothesis

We formulate the investigation as a deterministic numerical verification with the following decision framework.<sup>7</sup>

**$H_0$  (Null hypothesis).** The Riemann Hypothesis holds. All DQPTs in both System 1 and System 2 occur exclusively at  $\beta = 1/2$ . Equivalently, for all  $t \in \mathbb{R}$  and all  $\beta \in (0, 1)$  with  $\beta \neq 1/2$ :  $L(\beta, t) \neq 0$  and  $G(\beta, t)$  does not exhibit a zero corresponding to a zeta zero.

**$H_1$  (Alternative hypothesis).** The Riemann Hypothesis is false. There exists at least one DQPT at some  $\beta_0 \neq 1/2$ , corresponding to a nontrivial zero of  $\zeta(s)$  with  $\text{Re}(s) = \beta_0 \neq 1/2$ .

We note that failure to reject  $H_0$  does not constitute proof of RH; it merely provides additional numerical evidence consistent with RH up to the scanned range.

### 4.2 Test Protocol

The investigation proceeds in four stages.

**Stage 1: Implementation Verification.** Compute  $L(1/2, t_k)$  and  $G(1/2, t_k)$  at the first 100 known nontrivial zeta zeros ( $t_1 = 14.134725\dots$ ,  $t_2 = 21.022040\dots$ , etc.) and verify that they vanish within numerical tolerance. This confirms that the implementation correctly reproduces the theoretical predictions.

**Stage 2: Critical-Line Zero Detection.** Scan  $t \in [0, T]$  at fixed  $\beta = 1/2$ , detecting zeros of  $G(1/2, t)$  by sign changes. Compare the detected zero positions with published tables [3, 19]. Verify that the zero count matches the Riemann–von Mangoldt prediction  $N(T)$ .

**Stage 3: Off-Critical-Line Scan (System 1 only).** For each  $\beta \in [0.1, 0.9]$  with spacing  $\Delta\beta = 0.005$ , excluding  $[0.4975, 0.5025]$ , scan  $t \in [0, T]$  and compute  $|L(\beta, t)|$  using System 1 (the eta function). System 2 (the  $G$  function) is *not* used for off-line scanning because it can produce false positives at  $\beta \neq 1/2$  due to phase-alignment zeros (see Section 2.2.2). Record the minimum value of  $|L(\beta, t)|$  over the  $t$ -range

---

<sup>7</sup>Unlike a classical Neyman–Pearson test, there is no randomness here—the computation is deterministic. We adopt the hypothesis language for clarity of exposition, not to claim statistical inference.

for each  $\beta$ . If this minimum is below the zero-detection threshold  $\varepsilon$ , flag it for further investigation (Stage 4). Apply the adaptive refinement strategy described in Section 4.3 (Grid resolution).

**Stage 4: High-Precision Refinement.** For any candidate  $(\beta, t)$  flagged in Stage 3 with  $|L(\beta, t)| < \varepsilon$ :

- (a) Increase  $N$  and recompute to distinguish a genuine zero from a near-miss.
- (b) Refine  $(\beta, t)$  by 2D Newton's method or simplex optimization.
- (c) If the refined  $\beta$  is within tolerance of  $1/2$ , classify as a critical-line zero (consistent with  $H_0$ ). If  $\beta$  remains bounded away from  $1/2$ , classify as a potential counterexample to RH (rejecting  $H_0$ ).

### 4.3 Parameters

**Truncation parameter  $N$ .**

- *Phase 1 (verification):*  $N = 10^4$  with Euler–Maclaurin correction (5 terms). Justification: effective precision  $\sim 10^{-38}$ , far exceeding double-precision limits.
- *Phase 3 (scanning):*  $N = 10^3$  to  $10^4$ , depending on available compute. Justification: for a coarse scan, precision  $\sim 10^{-2}$  suffices to detect zeros; flagged candidates are refined in Phase 4.
- *Phase 4 (refinement):*  $N = 10^5$  with Euler–Maclaurin correction.

**Grid resolution.**

- $\Delta\beta = 0.005$  for the coarse scan (at least 179 off-critical-line  $\beta$  values in  $[0.1, 0.9]$  excluding a neighborhood of  $0.5$ ). The finer grid (compared to the naive  $\Delta\beta = 0.05$ ) is necessary because a zero at  $\beta_0 = 0.52$  would produce  $|\eta(0.50 + it_0)| \sim 0.02$ , which is too small to reliably detect at a grid point  $0.02$  away. With  $\Delta\beta = 0.005$ , such a zero would be sampled at  $\beta = 0.52$  (or within  $0.0025$  of it), where  $|\eta|$  is detectably small.
- $\Delta\beta = 0.001$  for refined scans near any flagged candidate.
- $\Delta t = 0.1$  for  $t \in [0, 1000]$  ( $10^4$  points per  $\beta$  value).
- $\Delta t = 0.01$  for refined scans near flagged candidates.

**Adaptive refinement strategy.** Zeros with  $|\beta_0 - 1/2| < \delta_{\text{threshold}}$  may produce only marginal dips in  $|L(\beta, t)|$  at nearby grid points. To address this:

1. *Initial coarse scan:* Evaluate  $|L(\beta, t)|$  on the  $\Delta\beta = 0.005$  grid and record all local minima below a liberal threshold ( $\varepsilon_{\text{coarse}} = 10^{-3}$ ).
2. *Adaptive refinement:* For each flagged  $(\beta, t)$  region, bisect in both  $\beta$  and  $t$  to find the true minimum of  $|L|$ . If the minimum decreases with refinement (indicating a genuine zero rather than a near-miss), continue refining until  $|L| < 10^{-6}$  or the minimum stabilizes.
3. *Convergence check:* Increase  $N$  and verify that  $|L|$  continues to decrease (as expected for a genuine zero) rather than stabilizing (indicating a near-miss).

#### Zero-detection tolerance.

- $\varepsilon = 10^{-6}$  for classifying a point as a zero (coarse scan).
- $\varepsilon = 10^{-12}$  for refined confirmation.

#### Scan range.

- $T_{\text{max}} = 1000$  for the initial investigation (capturing  $\sim 649$  zeros on the critical line).
- $T_{\text{max}} = 10,000$  for extended runs ( $\sim 10,142$  zeros).
- $T_{\text{max}} = 100,000$  as an aspirational target with GPU acceleration ( $\sim 138,069$  zeros).

#### Hardware.

- CPU: specified at runtime; any modern x86-64 processor suffices.
- GPU: NVIDIA RTX 2060 (1920 CUDA cores, 6 GB GDDR6 VRAM, Compute Capability 7.5).
- Precision: IEEE 754 double precision (64-bit,  $\sim 15.9$  significant decimal digits).

## 4.4 Validation

### 4.4.1 Unit Tests Against Reference Values

Each core mathematical function is tested against high-precision reference values computed independently using `mpmath` (Python arbitrary-precision library) or Mathematica. Specifically:

- $S_N(s)$  for  $N = 100$  and  $s = 0.5 + 14.134725i$ : compare to `mpmath`.
- $\vartheta(t)$  for  $t = 14.134725, 21.022040, 100, 1000$ : compare to `mpmath`.
- $\eta(s)$  for  $s = 0.5 + 14.134725i$ : compare to `mpmath`.
- $Z_H(t)$  for  $t = 14.134725$  (first zero): should be zero within precision.

Tolerance: agreement to 10 significant digits (limited by double precision and  $N$ ).

### 4.4.2 Cross-Validation: System 1 vs System 2

Both systems must yield the same zero positions (to within numerical precision) at  $\beta = 1/2$ . Specifically, if  $L(1/2, t^*) = 0$ , then  $G(1/2, t^*)$  must also change sign near  $t = t^*$ . We verify this for all detected zeros.

Tolerance:  $|t_L^* - t_G^*| < 10^{-4}$  for each corresponding zero pair.

### 4.4.3 Convergence Tests

For each of several fixed  $(\beta, t)$  points, compute  $L(\beta, t)$  for  $N = 10^2, 10^3, 10^4, 10^5$  and verify that the values converge. Specifically:

$$|L_N(\beta, t) - L_{N/10}(\beta, t)| < C \cdot (N/10)^{-\beta}, \quad (4.1)$$

where  $C$  is a constant. The convergence rate should be consistent with the theoretical  $O(N^{-\beta})$ .

### 4.4.4 Comparison with Odlyzko's Tables

The first 100 detected zeros are compared with Odlyzko's published tables (available from the LMFDB [19] and Andrew Odlyzko's website [3, 4]). Agreement is required to within the precision of our computation.

## 4.5 Controls

The following consistency checks serve as controls to verify the correctness of the implementation.

**Control 1.** At known zeros ( $t = t_k$ ), verify:

$$F_1(1/2, t_k) = 1 \pm \varepsilon, \quad (4.2)$$

where  $\varepsilon = 0.05$  (accounting for finite- $N$  effects).<sup>8</sup>

**Control 2.** Away from zeros, for generic  $t$  (e.g.,  $t = 50.0$ , which is not near any zero), verify:

$$F_1(\beta, t) = (1 - \beta) \pm \varepsilon \quad (4.3)$$

for  $\beta \in \{0.3, 0.5, 0.7\}$ .<sup>9</sup>

**Control 3.** Verify the zero count:

$$|N_{\text{detected}}(T) - N_{\text{predicted}}(T)| \leq 1, \quad (4.4)$$

where  $N_{\text{predicted}}(T) = \text{round}((1/\pi)\vartheta(T) + 1)$ . The tolerance of 1 accounts for the  $S(T) = (1/\pi) \arg \zeta(1/2 + iT)$  fluctuation term, which can cause the actual zero count to differ from the smooth prediction by  $\pm 1$  at specific  $T$  values.

**Control 4.** Verify that  $G(1/2, t)$  is real-valued (imaginary part  $< 10^{-14}$ ) for all sampled  $t$  values, confirming the  $Z$ -function property.

**Control 5.** Verify the functional equation numerically: for selected  $s$  values, confirm that  $\zeta(s)$  and  $\zeta(1-s)$  satisfy the functional equation (1.2) within precision. (This tests the Riemann–Siegel theta function implementation indirectly.)

## 5 Error Analysis

### 5.1 Truncation Error (Finite $N$ )

The dominant systematic error arises from truncating the Dirichlet series at  $N$  terms.

**Without Euler–Maclaurin correction.**

$$|\eta_N(s) - \eta(s)| \leq C(s) (N+1)^{-\beta}, \quad (5.1)$$

where  $C(s) = O(1)$  for bounded  $|s|$ . For  $\beta = 0.5$  and  $N = 10^4$ : truncation error  $\sim 10^{-2}$ .

---

<sup>8</sup>Under the Wei et al. log-base-2 normalization, this value would be  $\ln 2 \approx 0.693$ ; we use the natural log normalization throughout this document, giving  $F_1 = \beta_0 + 1 - \beta = 1$  at  $\beta = \beta_0 = 1/2$ .

<sup>9</sup>Under Wei et al. normalization:  $(1 - \beta) \ln 2$ .

**With Euler–Maclaurin correction ( $p$  terms).**

$$|\eta_N^{\text{EM}}(s) - \eta(s)| \leq C_p |s|^{2p-1} N^{-(\beta+2p-1)}, \quad (5.2)$$

where the factor  $|s|^{2p-1}$  accounts for the rising factorial in the correction terms (see Section 3.1). For  $\beta = 0.5$ ,  $p = 5$ ,  $N = 10^4$ , and  $t = 100$ : truncation error  $\sim 10^9 \times 10^{-38} = 10^{-29}$ . See the precision table in Section 3.1 (Table 1) for additional parameter combinations.

This error is purely systematic and can be reduced by increasing  $N$  (at the cost of more computation) or by choosing  $p$  optimally for the given  $(N, t)$  pair.

## 5.2 Floating-Point Precision

IEEE 754 double-precision arithmetic provides approximately 15.9 significant decimal digits (53-bit mantissa). The key sources of floating-point error are:

**Summation error.** Accumulating  $N$  terms in floating point introduces rounding error of order  $N \varepsilon_{\text{mach}} \sim N \times 10^{-16}$ . For  $N = 10^4$ : rounding error  $\sim 10^{-12}$ . For  $N = 10^6$ : rounding error  $\sim 10^{-10}$ . This can be mitigated by Kahan compensated summation [14], which reduces the error to  $O(\varepsilon_{\text{mach}})$  independent of  $N$ .

**Trigonometric evaluation and the FP64 precision wall.**  $\cos(t \ln n)$  and  $\sin(t \ln n)$  are evaluated by the standard library, with error  $O(\varepsilon_{\text{mach}})$  per evaluation. For large arguments ( $t \ln n \gg 1$ ), the argument reduction step introduces additional error proportional to  $t \ln(n) \varepsilon_{\text{mach}}$ . This constitutes a fundamental precision wall for double-precision computation:

| $t$       | Argument $t \ln n$ (at $n = 10^4$ ) | Effective digits lost |
|-----------|-------------------------------------|-----------------------|
| $10^3$    | $\sim 10^4$                         | $\sim 4$              |
| $10^5$    | $\sim 10^6$                         | $\sim 6$              |
| $10^7$    | $\sim 10^8$                         | $\sim 8$              |
| $10^{10}$ | $\sim 10^{11}$                      | $\sim 11$             |

At  $t > 10^7$ , fewer than 8 significant digits remain in the trigonometric evaluation, degrading zero detection to tolerances worse than  $10^{-8}$ . At  $t > 10^{10}$ , the argument reduction error ( $\sim 10^{-5}$ ) exceeds the zero-detection threshold, rendering double-precision computation unreliable.

For scientifically rigorous computation at large heights ( $t > 10^7$ ), arbitrary-precision arithmetic (e.g., MPFR / GNU MP, or at minimum IEEE 754 quad precision via `_float128` or `long double` on x86) is required. The Riemann–Siegel formula (Section 3.2.1) partially mitigates this issue by requiring only  $M = O(\sqrt{t})$

terms (reducing the maximum argument to  $t \ln M \sim t \ln \sqrt{t} = (t/2) \ln t$ ), but the theta function evaluation  $\vartheta(t)$  itself involves arguments of order  $t$ , so the precision wall persists. Our current implementation is therefore limited to  $t < 10^7$  for reliable zero detection in double precision, with reduced reliability up to  $t \sim 10^{10}$ .

**Cancellation near zeros.** When  $L(\beta, t)$  is near zero, the partial sum involves extensive cancellation. The absolute value of  $L$  near a zero is  $O(N^{-\beta})$ , while individual terms have magnitude  $O(n^{-\beta})$  with the largest terms  $O(1)$ . Without compensated summation, the absolute error in the sum is  $O(N \varepsilon_{\text{mach}})$  (each of  $N$  additions contributes  $O(\varepsilon_{\text{mach}})$  rounding error). With Kahan summation, the absolute error is reduced to  $O(\varepsilon_{\text{mach}})$ , independent of  $N$ . The value of  $L$  near a zero is  $O(N^{-\beta})$ . The relative precision is therefore:

$$\text{Without Kahan: relative error} \sim \frac{N \varepsilon_{\text{mach}}}{N^{-\beta}} = N^{1+\beta} \varepsilon_{\text{mach}}, \quad (5.3)$$

$$\text{With Kahan: relative error} \sim \frac{\varepsilon_{\text{mach}}}{N^{-\beta}} = N^{\beta} \varepsilon_{\text{mach}}. \quad (5.4)$$

For  $N = 10^4$  and  $\beta = 0.5$ : without Kahan, relative precision  $\sim 10^6 \times 10^{-16} = 10^{-10}$ . With Kahan summation:  $\sim N^{\beta} \varepsilon_{\text{mach}} = 10^2 \times 10^{-16} = 10^{-14}$ .

This is adequate for zero detection with tolerance  $10^{-6}$ .

### 5.3 Underflow Guard for $\log |L_N|$

The rate function  $F_1(\beta, t) = -(1/\ln N) \ln |L_N(\beta, t)|$  involves taking the logarithm of  $|L_N|$ . In floating-point arithmetic,  $|L_N|$  can underflow to exactly zero (when  $\beta$  is large or at a near-zero of  $\eta$  with large  $N$ ), producing  $\log(0) = -\infty$  and corrupting downstream computations.

**Guard implementation.** We use:

$$F_1 = -\frac{1}{\ln N} \ln(\max(|L_N|, 10^{-300})). \quad (5.5)$$

The floor value  $10^{-300}$  is well above the IEEE 754 double-precision minimum sub-normal ( $\sim 5 \times 10^{-324}$ ) but far below any physically meaningful value. When  $|L_N| < 10^{-300}$ , the computed  $F_1$  is flagged as a sentinel value indicating numerical underflow, and the point is marked for investigation with higher precision or larger  $N$ .

Additionally, any  $(\beta, t)$  point where  $|L_N| = 0$  exactly (to double precision) is recorded separately and not included in the rate function analysis, since it may represent either a genuine zero (signaling a DQPT) or a floating-point artifact.

## 5.4 Grid Discretization Error

Zeros detected by bisection on a grid with spacing  $\Delta t$  have position error bounded by  $\Delta t/2$  in the initial detection, refined to arbitrary precision by iterative bisection. After  $k$  bisection iterations, the position error is  $\Delta t \cdot 2^{-(k+1)}$ . For  $\Delta t = 0.1$  and  $k = 30$  iterations: position error  $\sim 10^{-10}$ .

In the  $\beta$  direction, the grid spacing  $\Delta\beta$  determines our ability to localize off-critical-line zeros. A zero at  $\beta_0$  will be detected (in the sense that  $|L|$  dips below  $\varepsilon$ ) only if  $\beta_0$  is within  $\sim \Delta\beta$  of a grid point. For  $\Delta\beta = 0.05$ , we could miss a zero with  $\beta_0 = 0.52$  (only 0.02 from the critical line). However, the minimum of  $|L(\beta, t)|$  over  $t$  is a smooth function of  $\beta$ , so a zero at  $\beta_0 = 0.52$  would also produce a small value at the nearest grid point  $\beta = 0.50$  (the critical line)—making it detectable, though its  $\beta$ -position would initially be misattributed to 0.50. The refinement stage (Stage 4) would then resolve its true position.

## 5.5 False Positive and False Negative Rates

**False positives** (declaring a zero where none exists) arise from numerical artifacts such as:

- Two nearly-cancelling large terms producing a spuriously small sum.
- Floating-point underflow near a local minimum of  $|L|$ .

Mitigation: every detected zero is confirmed by (a) increasing  $N$ , (b) computing both  $L$  and  $G$ , and (c) verifying consistency with the Riemann–von Mangoldt zero count.

**False negatives** (missing a genuine zero) arise from:

- Grid spacing too coarse relative to the zero separation.
- Insufficient  $N$  causing the partial sum to not resolve the zero.

Mitigation: grid spacing is chosen to be at most 1/4 of the expected zero separation (Section 3.3.1). The zero count check (Section 4.5, Control 3) provides an independent confirmation that no zeros were missed.

# 6 Expected Results

## 6.1 If RH Holds (Expected Scenario)

Under the assumption that RH is true—supported by all prior numerical evidence and by classical zero-density estimates, which imply that 100% of zeros lie on the

critical line in a density sense<sup>10</sup>—we expect:

1. **System 1 ( $L$ ):**  $L(1/2, t)$  vanishes at each of the  $\sim 649$  nontrivial zeros with  $0 < \text{Im}(s) \leq 1000$ , with  $|L(1/2, t_k)| < \varepsilon$  for each known zero  $t_k$ . For  $\beta \neq 1/2$ ,  $|L(\beta, t)|$  remains bounded away from zero for all  $t \in [0, 1000]$ .
2. **System 2 ( $G$ ):**  $G(1/2, t)$  exhibits sign changes at exactly the same positions as the Hardy  $Z$ -function  $Z_H(t)$ , with the detected zero positions matching Odlyzko’s tables [3] to within  $10^{-4}$ . The zero count matches the Riemann–von Mangoldt prediction.
3. **Rate function:**  $F_1(1/2, t)$  exhibits sharp peaks (approaching 1, or  $\ln 2$  under the Wei et al. log-base-2 normalization) at each zero, with  $F_1(1/2, t) \approx 1/2$  (or  $(1/2) \ln 2$  under Wei et al. normalization) between zeros. The 2D phase diagram  $|L(\beta, t)|$  in the  $(\beta, t)$  plane shows dark lines (near-vanishing  $|L|$ ) exclusively at  $\beta = 1/2$ .
4. **Off-critical-line scan:** No values of  $|L(\beta, t)|$  below the threshold  $\varepsilon = 10^{-6}$  for any  $\beta \neq 1/2$  in the scanned range.

## 6.2 If RH Fails (Unexpected Scenario)

If a counterexample to RH exists within our scanned range, we would observe:

1. A value  $(\beta_0, t_0)$  with  $\beta_0 \neq 1/2$  where  $|L(\beta_0, t_0)| < \varepsilon$ .
2. Upon refinement, the zero would persist and sharpen ( $|L|$  decreasing with increasing  $N$ ).
3. The phase diagram would show an off-critical-line zero—a dark point or line at  $\beta \neq 1/2$ .
4. Direct evaluation of  $\zeta(\beta_0 + it_0)$  via independent methods (e.g., Borwein acceleration or Euler–Maclaurin) would confirm the zero.<sup>11</sup>

Such a discovery would require extraordinary care in verification: independent reimplementations, arbitrary-precision computation, and comparison with direct zeta evaluation. The burden of proof for disproving RH is immense, and any claimed counterexample would need to withstand intense scrutiny.

<sup>10</sup>This density result already followed from the prior  $O(T^{3/4+\varepsilon})$  bound on the number of off-line zeros [20], since  $T^{3/4}/N(T) \rightarrow 0$ . Guth and Maynard’s 2024 improvement [21] sharpens the exponent to  $O(T^{3/5+\varepsilon})$ , the first improvement in over 80 years, but the qualitative conclusion of 100% density was already known.

<sup>11</sup>System 2’s  $G$  cannot be used for off-line confirmation due to the false-positive issue described in Section 2.2.2.

### 6.3 Limitations

We emphasize the inherent limitations of the numerical approach:

1. **Cannot prove RH.** Even scanning up to  $T = 10^{10}$  and finding all zeros on the critical line does not prove that all zeros (of which there are infinitely many) lie on the critical line. Proof requires a fundamentally different (analytical or algebraic) approach.
2. **Finite precision (FP64 wall).** Our computations use IEEE 754 double-precision arithmetic ( $\sim 15.9$  decimal digits). This imposes two distinct limitations: (a) a zero at  $\beta_0 = 1/2 + 10^{-20}$  would be indistinguishable from  $\beta_0 = 1/2$ ; (b) for  $t > 10^7$ , trigonometric argument reduction degrades precision below 8 significant digits, rendering zero detection unreliable (see Section 5.2 for the full analysis). Extending the implementation to arbitrary-precision arithmetic (MPFR<sup>12</sup>) would extend the reliable range to arbitrarily large  $t$ , at the cost of proportionally increased computation time.
3. **Finite scan range.** Zeros with  $\text{Im}(s) > T_{\max}$  are not tested. Counterexamples to RH, if they exist, might occur only at astronomically large heights. The first known failure of Gram’s law occurs at Gram point  $g_{126}$ , and it is conceivable that the first off-critical-line zero (if any) could be at heights far beyond computational reach.
4. **Indirect detection.** We detect zeros of  $L$  and  $G$ , which are related to but not identical with  $\zeta$ . Any discrepancy between  $L$ -zeros and  $\zeta$ -zeros must be attributed to finite- $N$  effects rather than interpreted as a new mathematical result.

## 7 Relation to Prior Work

### 7.1 Odlyzko’s Numerical Verification

Andrew Odlyzko’s seminal computations [3, 4] focused on computing specific high zeros of  $\zeta$ —around the  $10^{20}$ th and  $10^{22}$ nd zeros—and large blocks of consecutive zeros near these heights, providing crucial statistical data on zero spacing and GUE pair correlation. The systematic verification that the first  $10^{13}$  nontrivial zeros of  $\zeta$

---

<sup>12</sup>The GNU MPFR library provides correctly-rounded multiple-precision floating-point computation with arbitrary mantissa width, at the cost of  $O(p \log p)$  per operation for  $p$ -bit precision. For  $p = 128$  (quad precision), the overhead is approximately 4–10 $\times$  relative to FP64; for  $p = 256$ , approximately 10–25 $\times$ . This is a tractable overhead for extending reliable computation to  $t \sim 10^{15}$  or beyond.

lie on the critical line was carried out by Gourdon [5]. Both Odlyzko’s and Gourdon’s methods use the Odlyzko–Schönhage algorithm, which evaluates  $\zeta(1/2+it)$  at many points simultaneously using the Riemann–Siegel formula and FFT-based polynomial evaluation, achieving  $O(T^{1/2+\varepsilon})$  complexity for all zeros up to height  $T$ . These remain the gold standard for numerical verification of RH.

Our work does not compete with Odlyzko’s in terms of the number of zeros verified. Instead, it provides:

- An alternative computational framework based on quantum physics.
- A cross-check using an entirely different mathematical representation (eta function and  $Z$ -function via DQPT order parameters rather than direct zeta evaluation).
- Visualization of the  $\beta$ -dependence (the 2D phase diagram), which is not part of the standard numerical verification workflow.

## 7.2 Berry–Keating Conjecture

Berry and Keating [6] conjectured that the Hilbert–Pólya operator for RH is related to the quantization of the classical Hamiltonian  $H = xp$ . Specifically, they proposed that the operator  $(xp + px)/2$ , suitably regularized, has eigenvalues related to the nontrivial zeta zeros. This connects RH to quantum chaos: the zeta zeros would be energy levels of a chaotic quantum system, and their statistics (Montgomery’s pair correlation conjecture [22], confirmed numerically by Odlyzko) match those of random matrix theory (GUE ensemble).

The Wei et al. construction [1] is distinct from Berry–Keating in that it does not seek an operator whose *spectrum* encodes the zeros, but rather constructs systems whose *dynamical phase transitions* (temporal non-analyticities) correspond to zeros. The spectrum of  $H_0 = \sum \ln(n) |n\rangle\langle n|$  is logarithmic by design, and the zeros emerge in the time domain rather than the energy domain.

## 7.3 Connes’ Operator-Algebraic Approach

Alain Connes [23] formulated a trace formula relating the zeros of  $\zeta$  to the spectrum of a certain operator on an adelic space, providing a noncommutative geometry framework for RH. This approach has led to deep structural insights but has not yet yielded a proof of RH. Our computational work is complementary: it provides numerical data that any future proof (whether through Connes’ framework or otherwise) must be consistent with.

## 7.4 Guth–Maynard 2024 Breakthrough

In 2024, Larry Guth and James Maynard [21] proved improved large-value estimates for Dirichlet polynomials, which imply that at most  $O(T^{3/5+\varepsilon})$  zeros of  $\zeta(s)$  with  $0 < \text{Im}(s) \leq T$  can lie off the critical line, improving the previous  $O(T^{3/4+\varepsilon})$  bound (Ingham [20]; refined by Huxley and others). Note that the qualitative conclusion—that 100% of zeros lie on the critical line in a density sense (i.e.,  $N_{\text{off}}(T)/N(T) \rightarrow 0$ )—already followed from the prior  $O(T^{3/4+\varepsilon})$  bound since  $T^{3/4}/N(T) \rightarrow 0$ . Guth–Maynard’s contribution is the improved quantitative exponent, which represents the first improvement to the zero density estimate in the critical strip in over 80 years.

Our numerical investigation probes the complementary question: among the specific zeros in our scanned range, are there any off the critical line? Guth–Maynard tells us that at most  $T^{3/5+\varepsilon}$  could be off-line up to height  $T$ ; our computation would detect any such zeros within  $[0, T_{\text{max}}]$ .

## 7.5 How Our Implementation Differs

Our implementation is, to our knowledge, the first public classical simulation of the Wei et al. DQPT framework [1]. It differs from all the above in:

1. **Physical framing.** We compute quantum-mechanical observables (coherence functions, Loschmidt echoes, rate functions) rather than directly evaluating  $\zeta$  or  $L$ -functions.
2. **2D scanning.** We scan the full  $(\beta, t)$  plane, not just the critical line.
3. **DQPT characterization.** We identify zeros through their physical manifestation as phase transitions (singularities in the rate function), not merely as zero-crossings.
4. **Dual systems.** We implement both System 1 (eta-based) and System 2 ( $Z$ -function-based), providing internal cross-validation.

## 8 Directions Toward Proof

As discussed in Section 1.4, the DQPT–RH equivalence is a reformulation rather than a proof mechanism. Nevertheless, the physical language suggests several directions that might, in principle, lead to analytical progress on RH. We outline these directions here, emphasizing that none currently constitutes a proof strategy; they are research programs that could benefit from the physical intuition provided by the DQPT framework.

## 8.1 Lee–Yang Mechanism for Alternating Series

The Lee–Yang theorem [8] states that for certain classes of partition functions (e.g., ferromagnetic Ising models), all zeros of the partition function  $Z(z)$  as a function of the fugacity  $z$  lie on the unit circle  $|z| = 1$ . The DQPT framework interprets the eta function as a partition function of a system with logarithmic spectrum and alternating (anti-ferromagnetic) couplings.

If one could establish a Lee–Yang-type theorem for the partition function

$$Z(s) = \sum_{n=1}^{\infty} (-1)^{n+1} n^{-s}, \quad (8.1)$$

proving that its zeros (in the  $s$ -plane) are confined to the line  $\operatorname{Re}(s) = 1/2$ , this would prove RH. The key challenge is that the Lee–Yang theorem in its classical form applies to polynomial partition functions with specific positivity conditions, which do not directly apply to the Dirichlet eta function. Extending Lee–Yang-type results to infinite series with complex coefficients is a major open problem in mathematical physics.

## 8.2 Energy–Entropy Balance at $\beta = 1/2$

In statistical mechanics, phase transitions occur when the free energy  $F = E - TS$  (energy minus temperature times entropy) has a non-analyticity. At the critical temperature  $T_c$  (equivalently, critical inverse temperature  $\beta_c$ ), the energy and entropy contributions to the free energy are precisely balanced.

In the DQPT framework, the critical inverse temperature is  $\beta_c = 1/2$ . The “energy” of the system is related to the Dirichlet series  $\sum n^{-s} \ln n$  (the derivative of the partition function), and the “entropy” is related to the combinatorial structure of the alternating series. The conjecture that  $\beta_c = 1/2$  is the *unique* critical point is equivalent to RH.

A proof strategy along these lines would need to show that the energy–entropy balance condition is satisfied only at  $\beta = 1/2$ , for all  $t$ . This requires understanding the analytic structure of the free energy in the complex  $\beta$ -plane, which brings the problem back to the original analytic number theory.

**The Kramers–Wannier duality analogy.** Perhaps the most promising physical mechanism for fixing the critical point at  $\beta = 1/2$  comes from duality. In the two-dimensional Ising model, the Kramers–Wannier duality maps the partition function at inverse temperature  $\beta$  to the partition function at a dual temperature  $\beta^*$ , with the relation  $\sinh(2\beta) \sinh(2\beta^*) = 1$ . The critical point is the self-dual point  $\beta_c = \beta_c^*$ , which uniquely determines  $T_c$ .

The functional equation of the Riemann zeta function plays a formally analogous role:

$$\xi(s) = \xi(1 - s). \quad (8.2)$$

This is a duality that maps  $s$  to  $1 - s$ , fixing the critical line  $\text{Re}(s) = 1/2$  as the self-dual locus. In the DQPT framework, this becomes: the rate function  $F_1(\beta, t)$  satisfies a duality  $F_1(\beta, t) \longleftrightarrow F_1(1 - \beta, t')$  (where the mapping  $t \rightarrow t'$  involves the Riemann–Siegel theta function). The critical point  $\beta_c = 1/2$  is the unique self-dual point, just as in the Ising model.

The key question is whether this duality can be promoted to a *proof* that all DQPTs occur at the self-dual point  $\beta = 1/2$ . In the Ising model, the Kramers–Wannier duality combined with the uniqueness of the phase transition (which follows from the Peierls argument) proves that the critical temperature is at the self-dual point. The analogue for the DQPT framework would require:

- (a) Proving a “uniqueness of the phase transition” theorem: that for each  $t$ , there is at most one value of  $\beta \in (0, 1)$  where  $F_1$  has a singularity.
- (b) Combining this with the duality  $\xi(s) = \xi(1 - s)$  to conclude that this unique critical  $\beta$  must be  $1/2$ .

Neither (a) nor (b) is established, but this duality-based approach represents the most physically motivated path from the DQPT framework to a proof of RH. This is, in our view, the central open question raised by the Wei et al. construction.

### 8.3 Universality of the DQPT Critical Point

In equilibrium statistical mechanics, universality refers to the phenomenon that critical exponents and scaling functions are independent of microscopic details. If the DQPT at  $\beta = 1/2$  exhibits universality in a suitable sense—meaning that perturbations of the logarithmic spectrum  $E_n = \ln(n)$  do not move the critical point away from  $\beta = 1/2$ —this would suggest a structural reason for RH.

Specifically, consider deformed spectra  $E_n = \ln(n) + \varepsilon f(n)$  for small  $\varepsilon$  and various perturbations  $f$ . If the DQPT critical points remain at  $\beta = 1/2$  for all sufficiently small perturbations (structural stability), this would provide evidence that  $\beta = 1/2$  is “locked in” by some symmetry or topological constraint. Investigating the stability of the DQPT under spectral perturbations is a concrete computational project that could be carried out within our framework.

## 8.4 Selberg Trace Formula Connection

The Selberg trace formula (Selberg, 1956) relates the spectrum of the Laplace–Beltrami operator on a compact hyperbolic surface  $\Gamma \backslash \mathbb{H}$  to the lengths of closed geodesics on the surface. The Selberg zeta function  $Z_\Gamma(s)$  is an entire function whose zeros encode both the eigenvalues of the Laplacian and the topology of the surface.

The analogy with the Riemann zeta function is deep: the Selberg zeta function satisfies a functional equation, has a Riemann Hypothesis analog (which is *proved* for compact surfaces), and its zeros are related to a self-adjoint operator (the Laplacian). If the Wei et al. DQPT construction could be realized on a hyperbolic surface—with the logarithmic spectrum  $E_n = \ln(n)$  arising naturally from the geodesic lengths—then the proven Selberg RH might provide leverage toward the classical RH.

The key obstacle is that the Selberg zeta function is a product over *primitive* closed geodesics, while the Riemann zeta function is a product over *primes*. Establishing a precise correspondence between geodesic lengths and prime numbers (beyond the formal analogy) remains a central open problem.

## 8.5 Random Matrix Universality from the DQPT Perspective

Montgomery’s pair correlation conjecture [22], supported by Odlyzko’s numerical evidence [3], asserts that the local statistics of zeta zeros match those of eigenvalues of random matrices from the Gaussian Unitary Ensemble (GUE). In the DQPT framework, this would correspond to the statement that the spacing statistics of DQPT times  $t_k$  follow GUE statistics.

**Integrability obstruction.** A crucial limitation must be acknowledged: the Hamiltonian  $H = \sigma_z \otimes H_0$  with diagonal  $H_0 = \sum \ln(n) |n\rangle\langle n|$  defines a prototypically *integrable* system (a Gaudin-magnet-type central spin model). The eigenstates are product states of the probe spin and bath energy eigenstates, with no level repulsion. Integrable quantum systems generically exhibit *Poisson* level statistics (uncorrelated eigenvalue spacings), not GUE statistics. Therefore, the DQPT quantum system described in this paper *cannot* provide a physical mechanism for the observed GUE statistics of zeta zeros through its own dynamics.

The GUE universality of the zeta zeros must instead arise from the number-theoretic structure of the Riemann zeta function itself (encoded in the logarithmic spectrum  $E_n = \ln n$  and the resulting Dirichlet series), not from the quantum dynamics of the probe-bath coupling. Any connection between the DQPT framework and random matrix theory would need to operate at the level of the mathematical

structure of the partition function, not through a quantum chaos mechanism in the physical system [6]. This remains an open problem.

## 9 Implementation Architecture

### 9.1 File Structure

The project is organized into four directories: **src/** (core computation), **test/** (validation), **gpu/** (CUDA kernels), and **data/** (reference tables).

**Core source files (src/).**

| File                                | Purpose                                                                                |
|-------------------------------------|----------------------------------------------------------------------------------------|
| <code>dirichlet.h/.cpp</code>       | Partial Dirichlet series $S_N(s)$ and $\eta_N(s)$                                      |
| <code>partition.h/.cpp</code>       | Partition function $Z(\beta) = \sum n^{-\beta}$ with precomputation                    |
| <code>theta.h/.cpp</code>           | Riemann-Siegel theta function $\vartheta(t)$ ; Stirling expansion with error control   |
| <code>system1.h/.cpp</code>         | $\mathcal{L}(\beta, t)$ : accumulated phase factor (System 1) with alternating series  |
| <code>system2.h/.cpp</code>         | $\mathcal{G}(\beta, t)$ : Loschmidt amplitude (System 2) with $Z$ -function connection |
| <code>rate_function.h/.cpp</code>   | Rate function $\mathcal{F}(\beta, t)$ with underflow guard and sentinel detection      |
| <code>zeros.h/.cpp</code>           | Zero detection: bisection, Gram points, Turing zero-count check                        |
| <code>euler_maclaurin.h/.cpp</code> | Euler-Maclaurin tail correction; Bernoulli number computation                          |
| <code>kahan.h</code>                | Kahan compensated summation                                                            |
| <code>main.cpp</code>               | Entry point; CLI argument parsing                                                      |
| <code>config.h</code>               | Compile-time and runtime configuration                                                 |

**Test suite (test/).**

| File                              | Purpose                                                       |
|-----------------------------------|---------------------------------------------------------------|
| <code>test_dirichlet.cpp</code>   | Unit tests for $S_N$ and $\eta_N$ against <code>mpmath</code> |
| <code>test_theta.cpp</code>       | Unit tests for $\vartheta(t)$                                 |
| <code>test_known_zeros.cpp</code> | Verification against first 100 known zeros                    |

| File                                   | Purpose                                        |
|----------------------------------------|------------------------------------------------|
| <code>test_convergence.cpp</code>      | Convergence rate $O(N^{-\beta})$ verification  |
| <code>test_off_line.cpp</code>         | Off-critical-line scan                         |
| <code>test_rate_function.cpp</code>    | Rate function value checks (Controls 1–2)      |
| <code>test_cross_validation.cpp</code> | System 1 vs System 2 consistency               |
| <code>reference_values.h</code>        | High-precision reference values from<br>mpmath |

**GPU kernels** (`gpu/`).

| File                           | Purpose                                      |
|--------------------------------|----------------------------------------------|
| <code>sweep.cu</code>          | CUDA kernel for $(\beta, t)$ grid evaluation |
| <code>gpu_dirichlet.cuh</code> | Device-side Dirichlet series                 |
| <code>gpu_config.cuh</code>    | GPU-specific configuration                   |

**Data and output.** `data/zeros_known.txt` contains the first 100+ known non-trivial zeta zeros (sourced from the LMFDB [19] and Odlyzko [3]). The `results/` directory is populated at runtime.

## 9.2 Language Choices

**Core computation: C++17.** Chosen for:

- Direct control over floating-point arithmetic (IEEE 754 compliance via compiler flags `-ffp-contract=off`, `-fno-fast-math`).
- Efficient memory management for large arrays.
- Compatibility with CUDA (which requires C++ host code).
- No garbage collection pauses during long computations.

**GPU acceleration: CUDA C++.** Chosen for:

- Native support on the target hardware (NVIDIA RTX 2060).
- Fine-grained control over thread/block organization.
- Access to hardware special function units for `sin/cos/exp`.

**Test validation: Python/mpmath (external).** Reference values are precomputed using `mpmath` with 50-digit precision and hardcoded into the C++ test suite. This removes any runtime dependency on Python.

### 9.3 Dependencies

The implementation is designed to minimize external dependencies:

- **C++ standard library** (C++17): containers, algorithms, I/O, `<cmath>`.
- **CUDA Toolkit** ( $\geq 11.0$ ): for GPU kernels. Optional; the code compiles and runs in CPU-only mode without CUDA.
- **No other external dependencies.** No Boost, no Eigen, no BLAS/LAPACK, no numerical libraries. All required mathematical functions (Stirling expansion, Kahan summation, Bernoulli numbers, bisection) are implemented from scratch for full transparency and auditability.

### 9.4 Build System

A single Makefile with the following targets:

| Command                 | Action                       |
|-------------------------|------------------------------|
| <code>make</code>       | Build CPU-only version       |
| <code>make gpu</code>   | Build with CUDA GPU support  |
| <code>make test</code>  | Build and run all unit tests |
| <code>make clean</code> | Remove build artifacts       |

Compiler flags:

- `-std=c++17 -O2 -Wall -Wextra -Wpedantic`
- `-ffp-contract=off` (prevent fused multiply-add, which can change results)
- `-fno-fast-math` (ensure IEEE 754 compliance)
- For CUDA: `-arch=sm_75` (Turing architecture, RTX 2060)

## 10 Success Criteria (Quantitative)

Each criterion is stated as a specific numerical test with a pass/fail threshold.

## 10.1 Verification Criteria (Must Pass)

**V1. Known zero recovery (System 1).** For each of the first 100 known non-trivial zeros  $t_k$  ( $k = 1, \dots, 100$ ):

$$|L(0.5, t_k)| < 10^{-4} \quad \text{with } N = 10^4 \text{ and Euler-Maclaurin correction.} \quad (10.1)$$

Pass: all 100. Fail: any one exceeds threshold.

**V2. Known zero recovery (System 2).** For each of the first 100 known zeros:

$$G(0.5, t_k) \text{ changes sign in an interval of width } < 10^{-3} \text{ centered at } t_k. \quad (10.2)$$

Pass: all 100 sign changes detected within tolerance. Fail: any missed or mislocated.

**V3. Zero count accuracy.** For  $T \in \{100, 500, 1000\}$ :

$$|N_{\text{detected}}(T) - N_{\text{predicted}}(T)| \leq 1, \quad (10.3)$$

where  $N_{\text{predicted}}(T) = \text{round}((1/\pi) \vartheta(T) + 1)$ . The tolerance of 1 accounts for the  $S(T) = (1/\pi) \arg \zeta(1/2 + iT)$  fluctuation term, which can cause the actual zero count to differ from the smooth prediction by  $\pm 1$  at specific  $T$  values.

Pass: agreement within tolerance for all three  $T$  values. Fail: discrepancy  $> 1$  for any  $T$ .

**V4. Rate function at zeros.** For the first 10 known zeros:

$$|F_1(0.5, t_k) - 1.0| < 0.1 \quad (10.4)$$

(natural log normalization; equivalent to  $|F_1 - \ln 2| < 0.07$  under Wei et al. log-base-2 normalization).

Pass: all 10 within tolerance. Fail: any one outside.

**V5. Rate function away from zeros.** For  $t = 50.0$  (not near any zero) and  $\beta \in \{0.3, 0.5, 0.7\}$ :

$$|F_1(\beta, 50.0) - (1 - \beta)| < 0.05 \quad (10.5)$$

(natural log normalization; equivalent to  $|F_1 - (1 - \beta) \ln 2|$  under Wei et al. normalization).

Pass: all three  $\beta$  values within tolerance. Fail: any one outside.

**V6. Cross-validation (System 1 vs System 2).** For all detected zeros, the positions from System 1 and System 2 agree:

$$|t_k^{(1)} - t_k^{(2)}| < 10^{-3}. \quad (10.6)$$

Pass: all zeros agree. Fail: any discrepancy exceeding threshold.

**V7. Convergence rate.** Two convergence tests are performed:

- (a) *At a known zero:*  $s = 0.5 + 14.134725i$  (first zeta zero). The sequence  $|\eta_N(s)|$  for  $N = 100, 1000, 10000$  must decrease as  $O(N^{-0.5})$ . The ratio  $|\eta_{10N}(s)|/|\eta_N(s)|$  should be approximately  $1/\sqrt{10} \approx 0.316$ .

Pass: ratio in  $[0.25, 0.40]$ . Fail: ratio outside  $[0.20, 0.50]$ .

- (b) *At a non-zero point:*  $s = 0.5 + 50.0i$  (not a zeta zero;  $\eta(s) \approx -0.85 + 0.36i$ ). The sequence  $\eta_N(s)$  must converge to the reference value  $\eta(s)$  (computed via Borwein acceleration). The relative error  $|\eta_N(s) - \eta(s)|/|\eta(s)|$  for  $N = 100, 1000, 10000$  must decrease as  $O(N^{-0.5})$ .

Pass: relative error at  $N = 10000$  is below  $10^{-2}$ . Fail: relative error above  $10^{-1}$ .

## 10.2 RH-Relevant Criteria

**R1. Off-critical-line absence of zeros.** For all  $\beta \in [0.1, 0.9]$  with spacing 0.005, excluding  $[0.4975, 0.5025]$ , and  $t \in [0, 1000]$ :

$$\min_t |L(\beta, t)| > 10^{-3}. \quad (10.7)$$

Pass: no  $\beta$  shows a near-zero. Fail: any  $\beta$  shows  $|L| < 10^{-3}$  (triggers Stage 4 investigation, but is not by itself a failure of the implementation).

**R2. Phase diagram confinement.** The 2D heat map of  $-\log |L(\beta, t)|$  in the  $(\beta, t)$  plane shows strong signal (large  $-\log |L|$ ) exclusively in the column  $\beta = 0.5$ , with no comparable signal at other  $\beta$  values.

Pass: visual and quantitative confirmation. Fail: comparable signal at  $\beta \neq 0.5$ .

# 11 Timeline and Milestones

## Phase 1: Core Functions and Unit Tests (Weeks 1–2)

### Deliverables.

- Implemented and tested:  $S_N(s)$ ,  $\eta_N(s)$ ,  $Z(\beta)$ ,  $\vartheta(t)$ , Kahan summation, Euler–Maclaurin correction.
- Unit test suite passing against `mpmath` reference values.
- Verified: 10-digit agreement for  $S_N$ ,  $\vartheta$ ,  $\eta$  at selected test points.

**Milestone.** `make test` passes all Phase 1 tests.

## Phase 2: System Implementation and Known-Zero Verification (Weeks 3–4)

**Deliverables.**

- Implemented:  $L(\beta, t)$  (System 1),  $G(\beta, t)$  (System 2),  $F(\beta, t)$  (rate function).
- Zero detection algorithm (bisection + Gram points + Turing check).
- Verification:  $L$  and  $G$  vanish at first 100 known zeros (criteria V1, V2).
- Zero count matches Riemann–von Mangoldt (criterion V3).
- Rate function values correct (criteria V4, V5).
- Cross-validation passes (criterion V6).
- Convergence rate verified (criterion V7).

**Milestone.** All verification criteria (V1–V7) pass.

## Phase 3: Full $\beta$ - $t$ Scan (Weeks 5–6)

**Deliverables.**

- CPU-based scan of the  $(\beta, t)$  plane with  $\Delta\beta = 0.005$ ,  $\Delta t = 0.1$ ,  $T_{\max} = 1000$ .
- Off-critical-line analysis (criterion R1).
- 2D phase diagram generation and analysis (criterion R2).
- Investigation and resolution of any flagged candidates from Stage 3/4 of the test protocol.

**Milestone.** Criteria R1 and R2 evaluated. No false off-line zeros after refinement.

## Phase 4: GPU Acceleration and Large-Scale Computation (Weeks 7–10)

**Deliverables.**

- CUDA kernel for  $(\beta, t)$  grid evaluation.
- Performance benchmarking on RTX 2060.

- Extended scan:  $T_{\max} = 10,000$  (capturing  $\sim 10,142$  zeros).
- Fine  $\beta$  grid:  $\Delta\beta = 0.001$  for refined scans near flagged candidates.
- High-resolution phase diagram.

**Milestone.** GPU scan completes in  $< 1$  hour for  $T_{\max} = 10,000$ . All zeros confirmed on critical line.

## Phase 5: Results Analysis and Documentation (Weeks 11–12)

### Deliverables.

- Summary of all detected zeros and their  $(\beta, t)$  coordinates.
- Statistical analysis: distribution of  $|L(\beta, t)|$  as a function of  $\beta$ .
- Phase diagrams at multiple resolutions.
- Comparison with published tables (number of matching zeros, position accuracy).
- Final report documenting results, methodology validation, and conclusions.

**Milestone.** All success criteria met. Results documented.

## References

- [1] S. Wei, Y. Zhai, Q. Lu, W. Yang, P. Gao, C. Wei, J. Song, F. Nori, T. Xin, and G. Long. The Riemann Hypothesis emerges in dynamical quantum phase transitions. *arXiv preprint*, 2025. <https://arxiv.org/abs/2511.11199>.
- [2] B. Riemann. Über die Anzahl der Primzahlen unter einer gegebenen Grösse. *Monatsberichte der Berliner Akademie*, November 1859. <https://www.claymath.org/library/historical/riemann/riemann.pdf>.
- [3] A. M. Odlyzko. The  $10^{20}$ -th zero of the Riemann zeta function and 175 million of its neighbors. AT&T Bell Laboratories preprint. <https://www-users.cse.umn.edu/~odlyzko/unpublished/zeta.10to20.1989.pdf>, 1989.

- [4] A. M. Odlyzko. The  $10^{22}$ -nd zero of the Riemann zeta function. In M. van Frankenhuysen and M. L. Lapidus, editors, *Dynamical, Spectral, and Arithmetic Zeta Functions*, volume 290 of *Contemporary Mathematics*, pages 139–144. American Mathematical Society, 2001. <https://www-users.cse.umn.edu/~odlyzko/doc/zeta.html>.
- [5] X. Gourdon. The  $10^{13}$  first zeros of the Riemann zeta function, and zeros computation at very large height. <http://numbers.computation.free.fr/Constants/Miscellaneous/zetazeros1e13-1e24.pdf>, 2004.
- [6] M. V. Berry and J. P. Keating. The Riemann zeros and eigenvalue asymptotics. *SIAM Review*, 41(2):236–266, 1999. <https://doi.org/10.1137/S0036144598347497>.
- [7] M. Heyl, A. Polkovnikov, and S. Kehrein. Dynamical quantum phase transitions in the transverse-field Ising model. *Physical Review Letters*, 110:135704, 2013. <https://doi.org/10.1103/PhysRevLett.110.135704>.
- [8] T. D. Lee and C. N. Yang. Statistical theory of equations of state and phase transitions. II. lattice gas and Ising model. *Physical Review*, 87:410–419, 1952. <https://doi.org/10.1103/PhysRev.87.410>.
- [9] M. E. Fisher. The nature of critical points. In *Lectures in Theoretical Physics*, volume VIIC, pages 1–159. University of Colorado Press, 1965.
- [10] H. M. Edwards. *Riemann’s Zeta Function*. Academic Press, 1974. Reprinted by Dover, 2001.
- [11] E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function*. Oxford University Press, 2nd edition, 1986. Revised by D. R. Heath-Brown.
- [12] P. Borwein. An efficient algorithm for the Riemann zeta function. In *Canadian Mathematical Society Conference Proceedings*, 2000. <https://www.cecm.sfu.ca/personal/pborwein/PAPERS/P155.pdf>.
- [13] D. H. Lehmer. Extended computation of the Riemann zeta-function. *Mathematika*, 3:102–108, 1956. <https://doi.org/10.1112/S0025579300001753>.
- [14] W. Kahan. Further remarks on reducing truncation errors. *Communications of the ACM*, 8(1):40, 1965. <https://doi.org/10.1145/363707.363723>.
- [15] G. H. Hardy. Sur les zéros de la fonction  $\zeta(s)$  de Riemann. *Comptes Rendus de l’Académie des Sciences, Paris*, 158:1012–1014, 1914.

- [16] W. Gabcke. *Neue Herleitung und explizite Restabschätzung der Riemann-Siegel-Formel*. PhD thesis, Georg-August-Universität Göttingen, 1979.
- [17] T. Trudgian. On the success and failure of Gram’s law and the Rosser rule. *Acta Arithmetica*, 148:225–256, 2011. <https://doi.org/10.4064/aa148-3-2>.
- [18] A. M. Turing. Some calculations of the Riemann zeta-function. *Proceedings of the London Mathematical Society (Third Series)*, 3:99–117, 1953. <https://doi.org/10.1112/plms/s3-3.1.99>.
- [19] LMFDB – the L-functions and modular forms database. <https://www.lmfdb.org/>, 2024.
- [20] A. E. Ingham. On the estimation of  $n(\sigma, t)$ . *Quarterly Journal of Mathematics*, 11:291–292, 1940. <https://doi.org/10.1093/qmath/os-11.1.291>.
- [21] L. Guth and J. Maynard. New large value estimates for Dirichlet polynomials. *arXiv preprint*, 2024. <https://arxiv.org/abs/2405.20552>.
- [22] H. L. Montgomery. The pair correlation of zeros of the zeta function. In *Analytic Number Theory*, volume 24 of *Proceedings of Symposia in Pure Mathematics*, pages 181–193. American Mathematical Society, 1973.
- [23] A. Connes. Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Mathematica (New Series)*, 5:29–106, 1999. <https://doi.org/10.1007/s000290050042>.